IT Key Metrics Data 2019: Executive Summary

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This research contains enterprise-level IT spending and staffing metrics, as well as business productivity ratios for 21 vertical industries, collected throughout 2018 from a global audience. It provides an overall summary of the Key Industry Measures from the IT Key Metrics Data report series.

Key Findings

- Average cross industry IT spending as a percent of revenue was 3.6% in 2018.
- Average cross industry IT spending as a percent of operating expenses was 4.7%.
- Average cross industry IT spending per employee was $14,538.
- Average cross industry increase in IT spending from 2017 to 2018 was 3.1%. The average projected increase from 2018 to 2019 is 2.7%.
- Average portion of IT spending/budgets classified as a capital expense remains at 25%.
- Average personnel related costs account for the largest percentage of IT spending/budgets at 36%. SaaS, IaaS and public cloud services collectively accounted for 8%, which was up from 5% in 2017.
- Cloud adoption has resulted in shifts in the distribution of IT spending/budgets as well. Over the past several years, the percentage of IT spending/budgets on data centers has decreased, and now accounts for just 17% of the total. Investments in application development and application support have collectively increased and now account for 43% of IT spending/budgets.
- Average IT full-time equivalents (FTEs) as a percent of total employees was 5.3%.
- Average portion of IT FTEs that were contractors was 23%.
- Average portion of IT Spending/Budget dedicated to “Run the Business” was 71%.
- Average portion of IT Spending/Budget coming from the formal IT Budget increased from 79% to 84% from 2016 to 2018. This increase occurred as anecdotally we see the influence of business management over IT spending increasing. This difference may be due to the fact that our surveys reference a very specific definition of IT spending and staffing. This definition does not include certain items that are only IT-enabled, such as digital advertising expenses.

Recommendations

- Use this research (or your Gartner ITBudget Tool comparison report) as a source of comparative data to assist CIOs and IT leaders with fact-based decisions related to investments, planning, budgeting, ongoing operational assumptions and identification of quantitative best practices.
- These measures should be considered in the creation of future-state (both short- and long-term) objectives to quantify IT planning assumptions and to better understand niche or industry competitive drivers, inhibitors, conditions and trends.
- Use of this information should be considered the beginning of an ongoing measurement program. Organizations should consider investing in customized, refined, prescriptive or in-depth benchmarking engagements on a recurring basis to support the budget cycle, or whenever making significant, fact-based IT or business decisions.

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Analysis

How Does Your IT Organization Stack Up Against the Competition?

IT and enterprise leaders are challenged constantly with dynamic and disruptive market conditions, wherein the organization is evolving and technology is changing. The 2019 edition of the Gartner IT Key Metrics Data (ITKMD) series provides insight into the latest industry trends to help enterprises change, make fact-based decisions and answer key questions similar to these:

- Are you measuring the alignment between business and IT?
- Are your staffing and investment levels competitive in infrastructure and operations?
- Are you measuring your technology performance?
- Can you prove the success of current and future IT investments?

Big and disruptive enterprise changes require fact-based decisions regarding IT investments, costs and risks. A critical evaluation of IT capabilities — past, present and future — is the cornerstone of delivering business value. In general, clients find their journeys with benchmarking are more successful by participating in surveys, and in effect, they "get better at benchmarking by doing benchmarking."

This research provides an overview of the key findings on spending and staffing trends from leading organizations around the world and also provides the current comparison data included in the Gartner ITBudget Tool (see gartner.com/itbudget).

IT Key Metrics Data Research Background

The Gartner ITKMD series of reports was established in 1995 to support strategic IT investment decisions, and today the annual publication delivers more than 3,000 metrics, across 90 documents and covers 21 different industries. Allowing you to rapidly identify high-level IT spending, staffing, technology and performance trends.

In an ongoing effort to study, analyze, evolve and improve enterprise performance, Gartner drives a number of initiatives to continuously capture IT data and information from the greater Gartner client and non-client community to support the growth of the database, the industry insight and the published IT metrics series. We invite you to participate in and contribute to the study to represent your vertical industry and region. The Gartner client community provides an exemplary window into the global IT community, and, therefore, your participation is essential to this publication series.

To contribute to Gartner ITKMD research, start a survey and represent your industry and region. Surveys are available at: gartner.com/surveys.

IT Key Metrics Data Key Industry Measures Overview

This research contains relevant cross-industry averages and ranges from a subset of metrics and prescriptive engagements available through Gartner Benchmark Analytics consulting-based

capabilities. While cross-industry averages are indicative of enterprise IT spending levels, actual spending will vary around these averages when considering the variations of unique competitive landscapes, niche vertical industry subsectors, business scale, and IT complexity and demand, which may be justified by specific enterprise needs. These factors typically drive the context of an IT cost or performance evaluation and often dictate long-term support requirements. Ultimately, business value IT spending and staffing data should be used as a high-level directional indicator and in the creation of planning assumptions — not viewed as a prescriptive benchmark in which significant budget decisions are made.

For detailed information and metrics specific to each of the listed ITKMD vertical industries, see Table 1 or review "IT Key Metrics Data 2019: Index of Published Documents and Metrics" for a comprehensive list of all available IT Key Metrics Data 2018 research.

Table 1. ITKMD 2019 Key Industry Measures: Vertical Industry Document Index

Document Title	Current Year RN#	Multiyear RN#
Executive Summary	G00375583	
Midsize Enterprise Executive Summary	G00375584	
Cross-Industry	G00375592	G00375593
Banking and Financial Services	G00375594	G00375595
Chemicals	G00375596	G00375597
Construction, Materials and Natural Resources	G00375598	G00375599
Consumer Products	G00375600	G00375601
Education	G00375602	G00375603
Energy	G00375604	G00375605
Food and Beverage Processing	G00375606	G00375607
Government — National and International	G00375608	G00375609
Government — State and Local	G00375610	G00375611
Healthcare Providers	G00375612	G00375613
Industrial Electronics and Electrical Equipment	G00375614	G00375615
Industrial Manufacturing	G00375616	G00375617
Insurance	G00375618	G00375619
Media and Entertainment	G00375620	G00375621
Pharmaceuticals, Life Sciences and Medical Products	G00375622	G00375623
Professional Services	G00375624	G00375625
Retail and Wholesale	G00375626	G00375627
Software Publishing and Internet Services	G00375628	G00375629
Telecommunications	G00375630	G00375631
Transportation	G00375632	G00375633

Document Title	Current Year RN#	Multiyear RN#
Utilities	G00375634	G00375635

Source: Gartner IT Key Metrics Data (December 2018)

Using This Research

This research was commissioned to help IT and enterprise leaders compare IT investment levels (operational and capital expenses) with standard industry categories (revenue, operating expense, and total employees). To ensure a like for like comparison to the Gartner metrics it is important to adhere to the data definitions, which can be found throughout this research.

As with any published data, many potential interpretations and analyses exist. The dataset represents a mix of organizations of different sizes and vertical industry segmentations. The industry-specific spending profiles published here represent key metrics data collected directly from CIOs, CTOs, IT leaders and practitioners with respect to their organization's IT investment levels and future IT budgets. Most IT organizations follow an annual IT budgeting process and adjust their budgets based on changing economic and business conditions. In many organizations, IT spending levels are reviewed and revised on a quarterly or even monthly basis. Therefore, published IT spending benchmarks represent a "snapshot in time," and do not necessarily indicate what enterprises will or have ultimately spent on IT in the coming year or in the past.

Although the published figures represent what Gartner calls a "stalking horse" (that is, a position resulting from analysis of data that represents trends and results), each organization should assess its own situation carefully, and should not arbitrarily change to conform to published results (which do not necessarily represent best practices). For example, the metric of IT spending as a percent of revenue does not, by itself, provide valid comparative information that should be used to allocate IT or business resources. Moreover, IT spending statistics alone do not measure IT effectiveness and are not a gauge of successful IT organizations. They simply provide an indicative view of global investment levels for the market in general.

While the industry-specific spending metrics published here and in other research provide a high-level overview of spending priorities, many organizations feel the need to further evaluate their organization as compared to their niche competitive landscape when benchmarking. Many firms decide that a formal benchmarking exercise — one that is highly customized and prescriptive for the individual firm — is a natural follow-on to using the results presented in this research and in Gartner ITKMD publications. In such exercises, companies can be more assured that they are getting an "apples to apples" benchmark with a more refined peer group, and that the benchmark takes into consideration variations in complexity (such as the elements of industry, enterprise size, platforms, applications and other key variables).

Gartner recommends that organizations consider an investment in such customized or in-depth benchmarking engagements to support the budget cycle, significant IT or enterprise changes, or whenever making significant IT cost-based decisions. The information published in this research can be used during the time periods between prescriptive or consulting-led benchmark engagements.

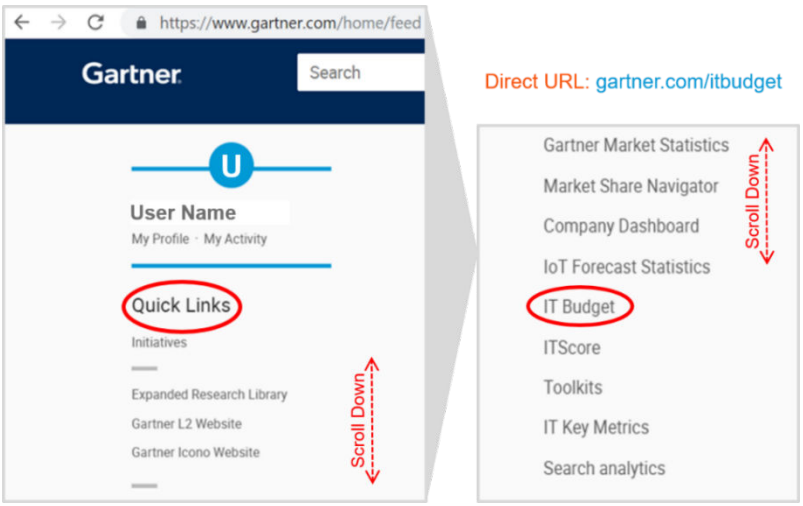
To explore Gartner's consulting-based prescriptive benchmark analytics capabilities, see the Appendix.

Gartner's ITBudget Tool: Compare Your IT Metrics to Your Industry

As an easier way to access the information in this report, you can use the Gartner ITBudget Tool to start your benchmarking journey and compare your enterprise IT metrics (see gartner.com/itbudget). The tool can be used on an ongoing basis, to analysis your current and future expenditure.

To start a new assessment and generate an IT metrics comparison report versus your industry, go to www.gartner.com, then scroll down and look at the left hand-side panel. Then click on **"IT Budget"** link. Finally select **"Start a New Assessment."**

Figure 1. ITBudget Tool Location



Source: Gartner (December 2018)

Note: Many CIOs and IT leaders leverage the "Delegation" feature to email data collection and financial alignment activities to a colleague to complete on their behalf; as well as to drive a common measurement reporting structure across independent divisions, agencies, or business units to support coordinated budgeting, planning and communication exercises. Delegates do not need to have access to Gartner.

Once you have completed an assessment, the following notes, along with those in the Gartner Recommended Reading, will highlight next steps to support IT business value discussions through IT financial transparency and cost optimization initiatives.

- "IT Key Metrics Data 2019: Resources to Review Your ITBudget Comparison Report"
- "Toolkit: Use Gartner's 2018 IT Key Metrics Data to Optimize IT Costs"

Access to the ITBudget Tool is dependent on your level of Gartner subscription.

Gartner IT Key Metrics Data Series

Depending on your subscription level for Gartner services, some clients have access to the complete Gartner ITKMD publication series. To access the series from gartner.com, select "Explore," "Metrics & Tools," and "IT Key Metrics Data."

ITKMD is part of the Gartner Benchmark Analytics range of solutions and offers a macro level look at Gartner's global database of comprehensive cost and performance measures. ITKMD provides you with immediate access to authoritative data on IT staffing and investment levels, as well as key technology cost and performance metrics. These metrics enable improved budget and investment decisions with regard to the changing environments of business and IT.

The ITKMD annual publication series contains more than 3,000 IT metrics published by way of 90 Gartner Benchmark Analytics research notes. In addition to the key IT financial metrics in this research, a variety of IT staffing and productivity metrics are available in the areas listed below. Some reports show vertical industry tendencies, while others tend to be cross-industry perspectives. Many of the metrics show averages by revenue scale or size of IT infrastructure environment supported (e.g., number of server operating system instances, number of installed MIPS, and number of personal computing devices).

These key metrics reports are broadly defined by five key areas of the IT portfolio:

- **Key Industry Measures.** Enterprise-level total IT spending and staffing metrics across 21 vertical industries, including current-year and multiyear averages. Metrics based on enterprise size in terms of annual revenues are often provided.
- **Key Infrastructure Measures.** IT functional area-specific unit cost, productivity and performance measures for the IT infrastructure environments, including current-year and multiyear averages for the Windows server, Linux x86 server, Unix server, mainframe, storage, end-user computing, IT service desk, data and voice network environments. Metrics by workload size are often provided.
- **Key Applications Measures.** Application development and application support spending and staffing metrics, project measures, life cycle phases, productivity and quality measures (current year and multiyear).
- **Key IT Security Measures.** Enterprise-level total spending and staffing measures.
- **Key Outsourcing Measures.** Enterprise-level total spending and staffing measures.

For a complete outline of all related published research in the series, see "IT Key Metrics Data 2019: Index of Published Documents and Metrics."

IT Key Metrics Data Source

Information for ITKMD is continuously collected worldwide via direct fact-finding in our many benchmarking and consulting engagements, through surveys of the Gartner community and at Gartner events, in addition to surveys of non-Gartner-based communities. Financial information,

such as revenue and operating income, is also collected from secondary research sources, such as annual reports and public databases.

Data Variations

As information for ITKMD is compiled by Gartner from multiple sources, we do not use a specific sampling method. The data collected each year may have a different distribution of organization sizes (revenue/business operating expenses/employees), and geographies. While we do group similar companies within each of the individual industry categories, there is always some diversity in businesses represented. For this reason, there may be minor or significant fluctuations in metrics from year to year.

This IT spending and staffing report contains 2019 projections for the IT spending as a percent of revenue, IT spending as a percent of operating expenses, and IT spending per employee metrics. Only some of our survey participants provided projections for 2019. Therefore, the sample size available for projections is smaller than the sample sizes for current and previous years. For this reason, the 2018 metrics may be more appropriate to use as a comparison than those of 2019. The 2018 metrics will also match the information published in our ITKMD current-year reports.

Demographics

In 2018, Gartner collected 5,311 data points in total from public and private enterprises from more than 80 countries in 21 industry sectors to contribute towards all of the IT Key Metrics Data series of reports. For more information, including the distribution of data points by region, see "IT Key Metrics Data 2019: Demographics."

For the key industry measures contained in this report, we collected 2,735 data points. The result is the most comprehensive and authoritative IT spending, staffing and performance data in the industry.

Table 2 shows the number of observations and average size of the organizations that represented in each industry (annual revenue and number of employees).

Table 2. Number of Observations, Average Revenue and Enterprise Employees

Industry	Number of Observations	2017 Revenue (Billions of USD)	2018 Employees (Thousands)
All Industries (Cross-Industry)	2,735	4.7	13.9
Banking and Financial Services	327	3.5	10.1
Chemicals	55	4.7	9.2
Construction, Materials and Natural Resources	146	3.5	9.5
Consumer Products	95	3.3	12.7
Education	163	0.8	5.4
Energy	64	13.1	5.8
Food and Beverage Processing	115	6.0	17.6
Government — National/International (Operating Budget)	156	10.9	32.3
Government — State/Local (Operating Budget)	167	1.8	8.6
Healthcare Providers	104	2.1	13.2
Industrial Electronics and Electrical Equipment	73	8.6	25.6
Industrial Manufacturing	160	8.2	23.4
Insurance	219	5.4	5.3
Media and Entertainment	52	1.4	5.4
Pharmaceuticals, Life Sciences and Medical Products	84	6.7	14.8
Professional Services	215	2.1	10.4
Retail and Wholesale	199	8.4	27.2
Software Publishing and Internet Services	44	1.9	4.9
Telecommunications	37	7.7	17.2
Transportation	124	4.5	22.5

Industry	Number of Observations	2017 Revenue (Billions of USD)	2018 Employees (Thousands)
Utilities	136	3.8	5.4
Notes: (1) The revenue figures reported are final and official for 2017; the 2018 revenue figures were not announced or were otherwise unavailable at the time of this publication. (2) Government operating budget is used as a proxy for "revenue"; however, it is not included in the all-industry average for revenue. (3) The all-industry enterprise full-time equivalent (FTE) average includes government FTEs.			

Source: Gartner IT Key Metrics Data (December 2018)

For further details on the industry definitions see report "IT Key Metrics Data 2019: Definition of Industries."

2018 IT Investment Measures

Total IT Spending/Budget Definition

For the purpose of this research, Gartner has defined "total IT spending" as the following:

The best estimate of total spending at the end of the 12-month budget period for IT to support the enterprise. IT spending/budget can come from anywhere in the enterprise that incurs IT costs, and it is not limited to the IT organization. It includes estimates by enterprises on decentralized IT spending and or "shadow IT." It is calculated on an annualized 'cash flow view' basis, and, therefore, contains capital spending and operational expenses, but not depreciation or amortization.

From a Resource (Asset) Perspective IT Spending/Budget Includes

- Hardware, software, personnel for IT FTEs including IT contractors, network transmission, outsourced IT services (e.g., consulting, system integration, infrastructure etc.), public cloud services, occupancy spending associated with IT personnel and data centers, taxes other than value added taxes where it its recovered and refunded to the organization.

From an IT Functional Area or Activity Perspective IT Spending/Budget Includes

- Data center, end-user computing, IT service desk, voice and data network, application development, application support, corporate IT management, finance and administration.

From a Business Perspective IT Spending/Budget Includes

- Business functionality software that runs on standard IT equipment e.g., SAP, insurance claims processing software.
- Dedicated general use data processing equipment used in operations, production and engineering environments. Examples are standard computing equipment used in computer-

aided design/computer-aided manufacturing (CAD/CAM) or for factory automation, and tablets used by healthcare professionals or market researchers performing surveys.

- Spending on hardware, analytics software, development tools, and staff who create and maintain the business analytics applications.

IT Spending/Budget Excludes

- Equipment that contains computerized components, but was built or purchased for non-data-processing purposes. Examples include robotic manufacturing machines, automated teller machines, specialized point-of-sale devices, scanners, blood pressure monitors and sensors used in Internet of Things (IoT) applications.
- Appliance-like or proprietary data processing equipment that has a single (typically industry vertical) purpose and cannot be used for other general purposes. An example of an appliance like device is a programmable logic controller on a supervisory control and data acquisition (SCADA) system. This device can only control the flow of electricity through the power grid. Since it cannot be repurposed, it is not included.
- Business data subscriptions and services (such as Bloomberg, S&P Capital IQ, Dun & Bradstreet), even if they are managed by the IT organization.
- Business process outsourcing services (BPO) where organizations outsource entire business functions such as payroll or benefits management. This includes cases where the BPO vendor provides access to software, and also guarantees that the outcomes of their services will meet business requirements, such as tax and withholding regulations. Note: where a vendor provides Software as a Service (SaaS) and only guarantees that the software will perform as specified, then this is in scope of the IT spending/budget.
- Personnel performing business analytics functions e.g., data scientists are other users of business analytics capabilities.
- Content Functions – e.g., gathering, and development of information to be published on a corporate intranet.

Notes For Providers of Information and Technology Services to External Customers

For the purposes of this definition, “commercial software” includes applications:

- That are resold as packaged software or public cloud services.
- That deliver third party digital content and/or analytics e.g., market data feeds, internet search, music streaming etc., social media platforms.
- That are embedded in other products e.g., automobiles.

For Providers of Information and Technology Services to External Customers IT Spending/ Budget Includes

- IT assets used by commercial software developers, e.g., development and testing tools.

- The IT based delivery mechanisms for commercial software. This may include servers hosting software to be downloaded, or infrastructure that runs SaaS applications and websites.
- Applications developed for customer use (web/mobile/other), primarily to enhance the value of other products and services sold e.g., insurance mobile application.
- Applications and technology developed to provide business process outsourcing capabilities for external clients. e.g., processing payroll.

For Providers of Information and Technology Services to External Customers IT Spending/ Budget Excludes

- IT-skilled employees who provide services for the organization's external clients.
- Spending for personnel responsible for developing commercial software (as these are considered users/employees not IT FTEs).
- Spending for technology that is used to provide infrastructure services that are sold e.g., carrier networks, hardware and software used in hosting or Infrastructure as a Service, or Platform as a Service.

Points of Clarification

The IT Key Metrics Data: Key Industry Measures publication series looks at IT spending from a "cash flow" view, in that IT spending is defined as "the total of the IT operating budget (excluding depreciation and amortization), plus IT capital expenditure for the current year." This view allows organizations to understand the current year's cash outlay based on current-year management, plans and future-state strategy objectives. Many organizations monitor IT spending results using what is sometimes called the annualized "book view" or "accounting view," which represents the IT operating expenditure budget, including current depreciation (the allocation of prior years' IT capitalized expenses, which the enterprise records on its books for the current year). Capital budgets for the year in this accounting-based view are typically collected and reported separately. While this accounting view is helpful in outlining the annual cost of IT, it often does not accurately reflect the current strategy (and the respective investment decisions) because depreciation represents decisions made in the past. In this research, the ratio of IT operating versus capital spending is provided so that detailed comparisons can be made.

It should be noted that IT spending as a percent of revenue in the ITKMD is calculated on the basis of the current year's IT spending (budget) divided by the previous year's stated revenue. The calculation is made in this way because the current year's financial data is typically not available at the time of publication, while the IT spending/budget data is available. Also, the IT budget for a future year is based on experience from the current year.

Although Gartner publishes worldwide vertical-industry-specific IT spending and staffing metrics, ITKMD does not publish metrics by vertical industries within key geographic regions because previous research has shown that spending patterns are broadly similar by vertical market across regions. So, for example, financial services tend to spend a relatively high percentage of revenue

compared with other vertical industries, whether the company is in Europe, Asia/Pacific or North America. While this information may not be available through published research, Gartner does provide prescriptive benchmarking services versus unique peer groups by industry subsector and or marketplace through the Gartner Consulting capabilities on a service for a fee basis.

To explore Gartner's consulting based prescriptive benchmark analytics capabilities, see the Appendix.

IT Intensity

IT intensity, which is defined as "the level of IT investment relative to business results," is a fundamental tool in business strategy and IT communications. Although many organizations focus on one measure to understand their relative IT investment levels, Gartner has found that no single measure tells the whole story, and that the metrics need a business context to drive value and meaning for the greater enterprise to leverage. Gartner suggests that clients view IT investment against multiple measures of business volume and financial performance, and then triangulate on these as compared with various ratios of business efficiency and productivity. Next, incorporate these metrics into regular communications with the appropriate business context to link IT investment to business performance indicators.

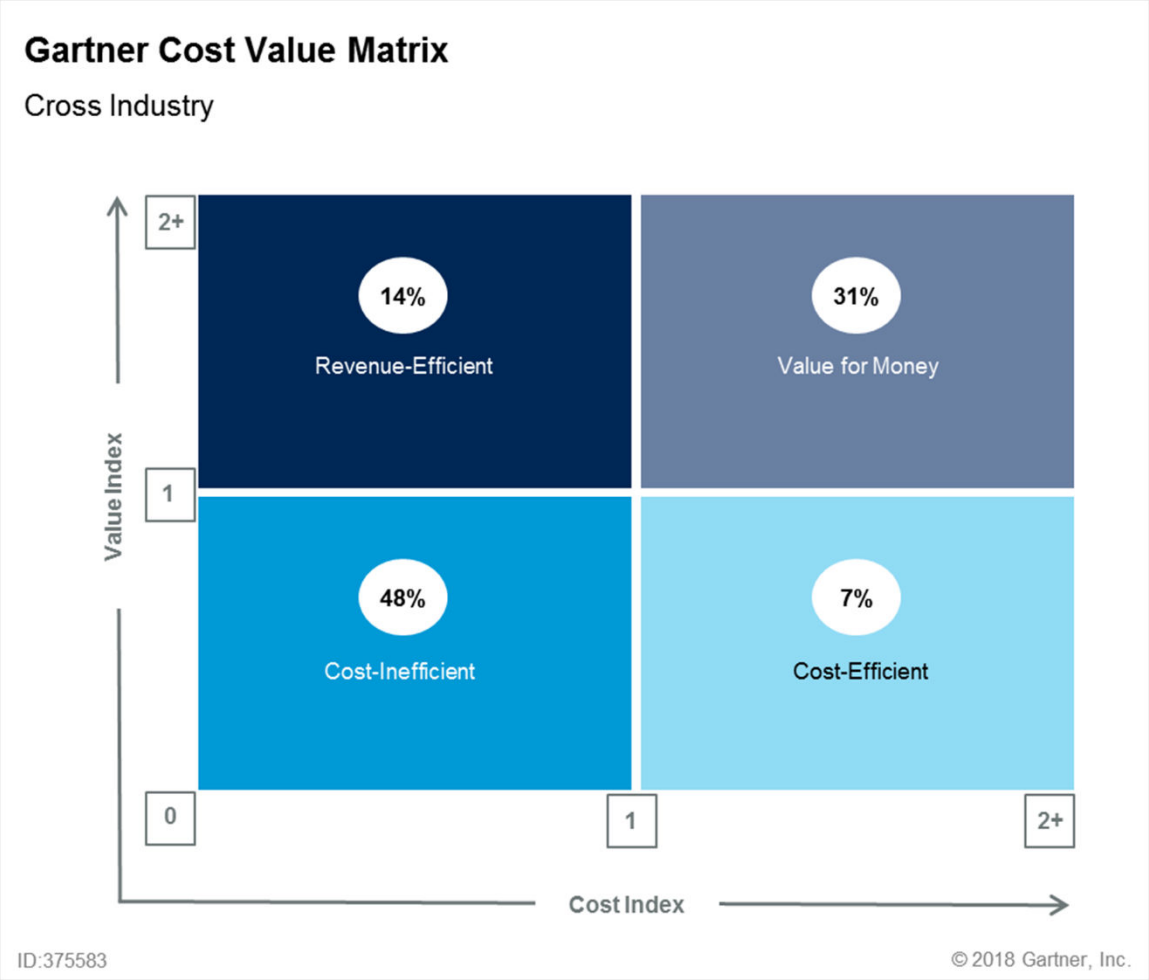
For the purpose of the IT Key Metrics Data: Key Industry Measures publication series, Gartner suggests that clients triangulate between IT spending as a percent of revenue, IT spending as a percent of operating expense, and IT spending per company employee to understand relative IT intensity levels. Next, clients should cross-examine those measures against business productivity ratios (such as revenue per employee, operating income per employee and profitability ratios) to understand the impact on the business. For more information on these metrics, see the individual industry-specific documents outlined in Table 1 above.

Gartner’s Cost Value Matrix: Finding the Right Balance between your IT Cost Optimization and IT Investment Efforts

CIOs often measure success of cost optimization plans by measuring total IT cost reduction. This can distract from the ultimate goal of helping to execute on business strategy. Gartner’s cost value matrix provides insight to ensure that cost optimization plans maximize business value as well as reduce costs.

Gartner developed the cost value matrix (CVM) to provide a better way to find the right balance between how much time/funding/resources you spend on IT cost optimization efforts versus investment in technology to optimize the business costs and drive top line growth through innovation in business models. See "The Gartner Cost Value Matrix: Assess If Cost Optimization Initiatives Balance Cost Reduction and Business Value."

Figure 2. Gartner Cost Value Matrix: Distribution of Enterprises by Quadrant



Source: Gartner IT Key Metrics Data (December 2018)

The Four Quadrants in the CVM Represent Different Combinations of Cost Index and Value Index Scores

- **Cost-Inefficient (Value Score Less Than 1, Cost Score Less Than 1):**

  Organizations in the "cost-inefficient" quadrant had a score of less than 1 on both the value and cost index. Most organizations with this score had higher-than-average IT spending, with a lower-than-average "revenue per employee" and "income per employee" metrics. In an efficient organization, we would expect the higher spending to be offset by having better-than-average business metrics. In the absence of such improvements, the conclusion is that there is a genuine overspend in the IT budget, which needs to be reduced in line with industry averages, or that that the organization is undergoing a period of investment in IT, which once finished will yield benefits both in terms of IT spending levels reaching a normal run rate and an improvement in business metrics. We encourage organizations to calculate their cost value

rating (CVR) for both the current state (as is) and future state (to be — a point in time where IT spending is back at normal levels and business metrics are showing benefits of investments) to ensure that the organization will move into the appropriate quadrant once the benefits of IT investment are realized. If both the current cost value matrix (CVM) rating and the future CVM rating give the organization a "cost-inefficient" status, then it is likely the IT budget has inefficient overspending and therefore should be bought back in line with industry averages. An IT cost reduction exercise would be recommended as a next step.

- **Revenue-Efficient (Value Score More Than 1, Cost Score Less Than 1):**

  IT organizations in this quadrant had a value score of greater than 1 and a cost score of less than 1, which means they generated a higher-than-average “revenue per employee” metric with a lower-than-average IT spend. Organizations in this quadrant use IT predominantly to support business growth rather than improve operational back-office processes (because they had lower-than-average business margins — that is, “income per employee”). These IT organizations seem to be extremely innovative and more front-office-focused because they are able to maximize the use of technology in order to drive additional revenue, enhance customer experience and reduce customer churn.

- **Value for Money (Value Score More Than 1, Cost Score More Than 1):**

  Both the cost and value scores are greater than 1. Organizations in this quadrant not only are generating a higher operating margin (income per employee) and “revenue per employee,” but also they are able to achieve this above-average business value at a reasonable cost compared with peers in our dataset. This should be the long term target of any IT organizations, and CIOs should create a roadmap to achieve or sustain the VFM status over the long term.

- **Cost-Efficient (Value Score Less Than 1, Cost Score More Than 1):**

  IT organizations in this quadrant have a high operating profit relative to their IT spend — that is, their value score is less than 1 and their cost score is higher than 1. This could be caused by several factors. CIOs in this group either benefit from significant economies of scale, which lead to below-average IT spend, or their primary focus is to optimize efficiency and operating margins instead of improving business metrics.

Note: Like any model the results are dependent on the precision of the inputs and the relevance of the measures. Some key things to keep in mind are:

- The standard industry peer group used here may not match your organization perfectly in terms of scale. The same issues you face in comparing your IT spending measures to Gartner’s standard industry categories will apply here.
- Your staffing strategy can affect results. For example, using a higher number of lower cost staff will drive down your revenue per employee, but will still have no discernable effect on revenue generation.
- The Cost Value Matrix framework evaluates business productivity in terms of revenue generation intensity; therefore, this framework is not applicable to public-sector or non-revenue-generating organizations.

With these caveats Gartner provides the following guidance in using CVM:

1. Focus on where you are today versus where you plan to be in the future. Consider what investment changes are required to move upward and/or to the right – improving efficiency and/or value rather than viewing the model as a “report card”.
2. Consider looking at metrics other than revenue per employee and operating income per employee to define value. For example an airline might examine revenue per passenger mile instead of revenue per employee and an insurance company might also analyze the combined ratio in addition to operating income per employee to illustrate business performance.

Recommended Reading

- "Use the Cost Value Matrix to Bring Balance and Direction to Cost Optimization Efforts"
- "The Gartner Cost Value Matrix: Assess If Cost Optimization Initiatives Balance Cost Reduction and Business Value"
- "The Gartner Cost Value Matrix: Insights From the First 1,500-Plus Assessments"

IT Spending as a Percent of Revenue, 2018

IT spending as a percent of revenue is the most recognized measure of total IT investment relative to top-line business results.

Revenue is defined as:

> "The enterprise revenue associated with the business units supported by the IT organization (banks should use total interest income plus noninterest income minus provision for loan losses, while insurance companies should use gross written premiums and other income)."

The value of this measure is that it assists in identifying the competitiveness of investment levels relative to the most fundamental measure of business performance: revenue. While this has been viewed as a must-have and readily available metric for many enterprises, common misuses include:

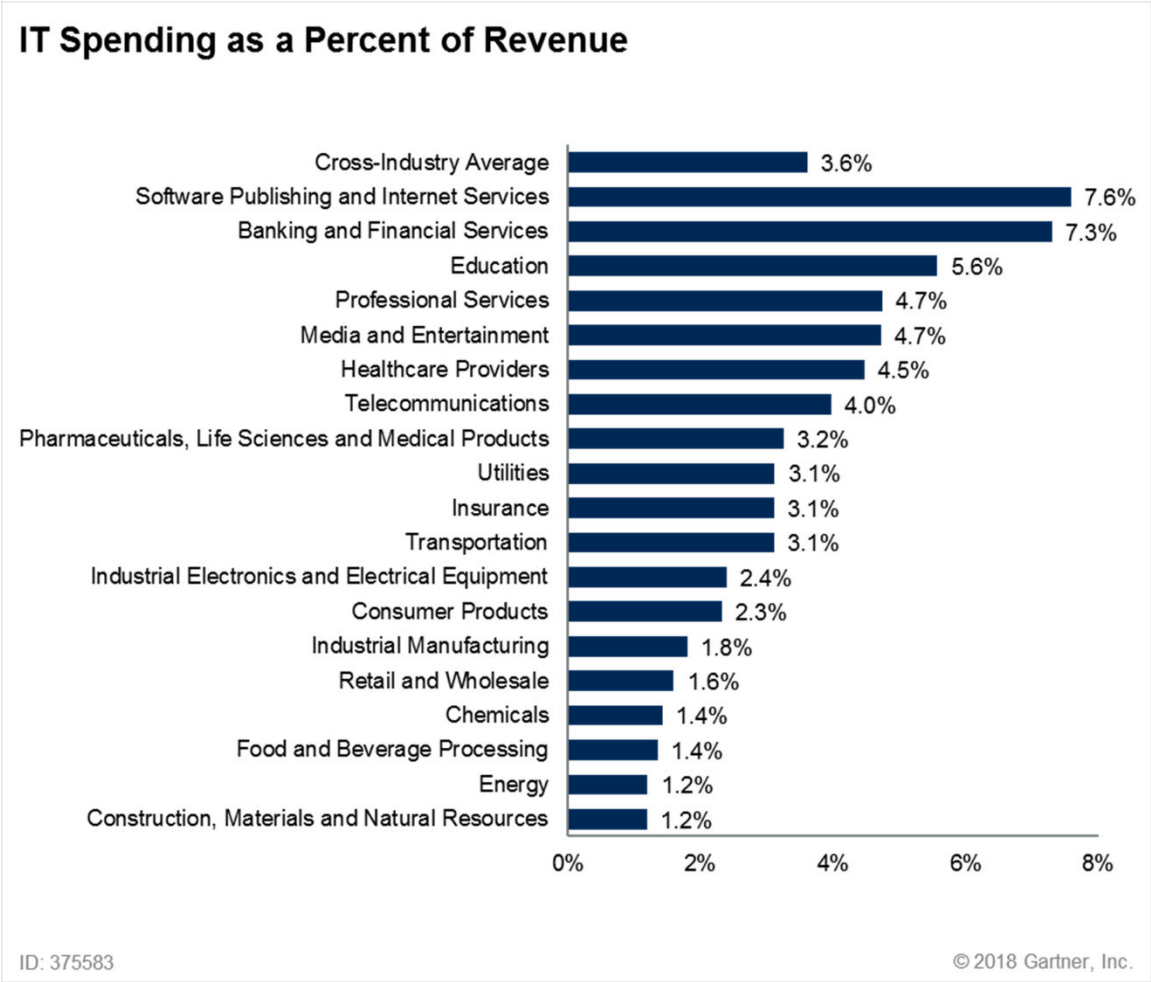
- Looking at a single year rather than multiyear trends
- Basing decisions on the assumption that this figure will not change in the future, sometimes dramatically
- Failing to understand and address changes in the numerator and the denominator of the calculation
- Considering just the average rather than the range of values or the upper and lower quartiles, which can be found in Gartner ITKMD reports specific to vertical industries (See individual industry-specific documents outlined in Table 1 above.)

IT spending as a percent of revenue alone does not highlight why spending levels are at, above or below average (which are often misinterpreted as "good" or "bad"), nor does it reflect IT's

contribution to business performance. Thus, IT spending as a percent of revenue needs to be considered in tandem with other IT intensity measures, as well as the context of business objectives, the rate of change and the overall circumstances affecting the numerator, as well as the denominator, of the calculation.

IT intensive industries like software publishing and internet services, and banking and financial services continue to top the list of IT spending as a percent of revenue. While more “non-IT capital” and labor-intensive industries like construction, materials and natural resources and food and beverage processing are at the bottom (see Figure 3).

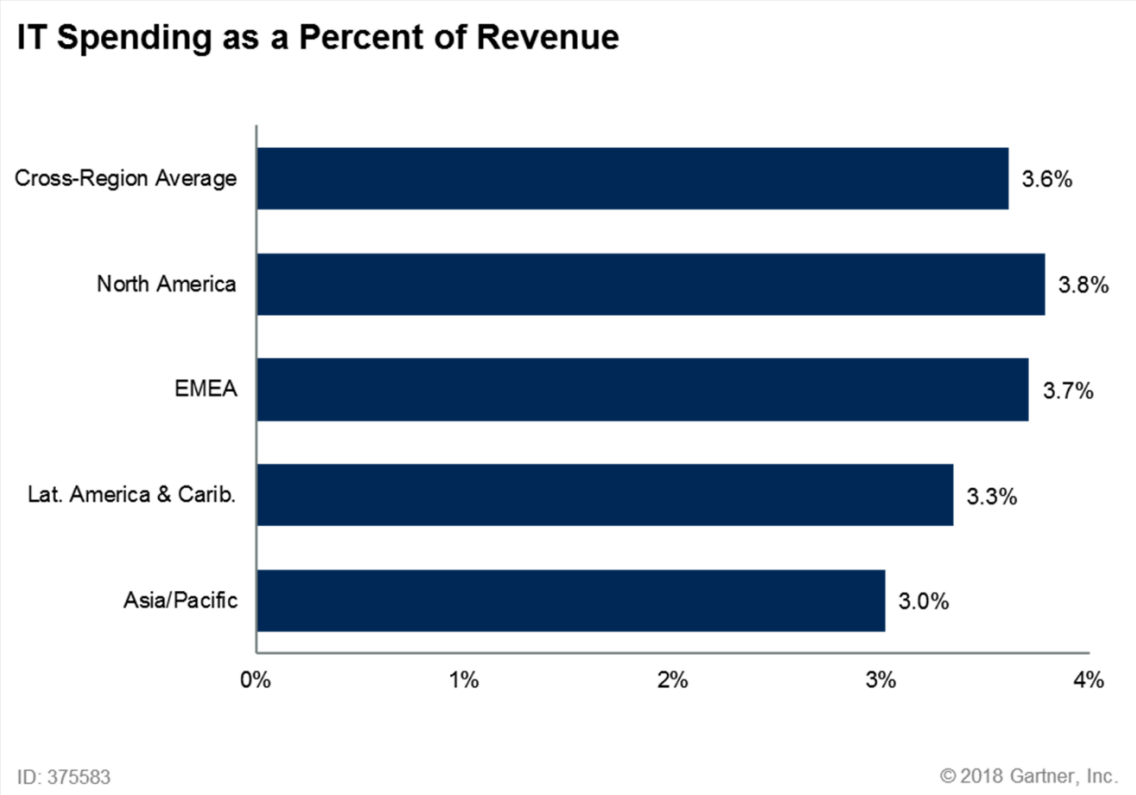
Figure 3. IT Spending as a Percent of Revenue, by Industry, 2018



Source: Gartner IT Key Metrics Data (December 2018)

Regionally, North America has the highest level of average IT spending as a percent of revenue, while Asia/Pacific has the lowest (Figure 4). One reason that North America and EMEA tend to be higher than Latin America and Asia/Pacific because they generally have a higher cost for personnel. Higher personnel costs result in more situations where it pays to invest in technology for automation to reduce overall costs.

Figure 4. IT Spending as a Percent of Revenue, by Region, 2018



Source: Gartner IT Key Metrics Data (December 2018)

Note that IT spending as a percent of revenue is calculated on the basis of the current year's IT spending, divided by the previous year's revenue. We make the calculation in this way because the IT budget for a future year is based on experience from the current year. However, for practical reasons, we use the previous year's revenue because the current year's financial information is not available to us at the same time as the IT spending numbers.

IT Spending as a Percent of Operating Expense, 2018

IT spending as a percent of operating expense is another view of IT investment levels in terms of the role IT plays in overall business spending patterns.

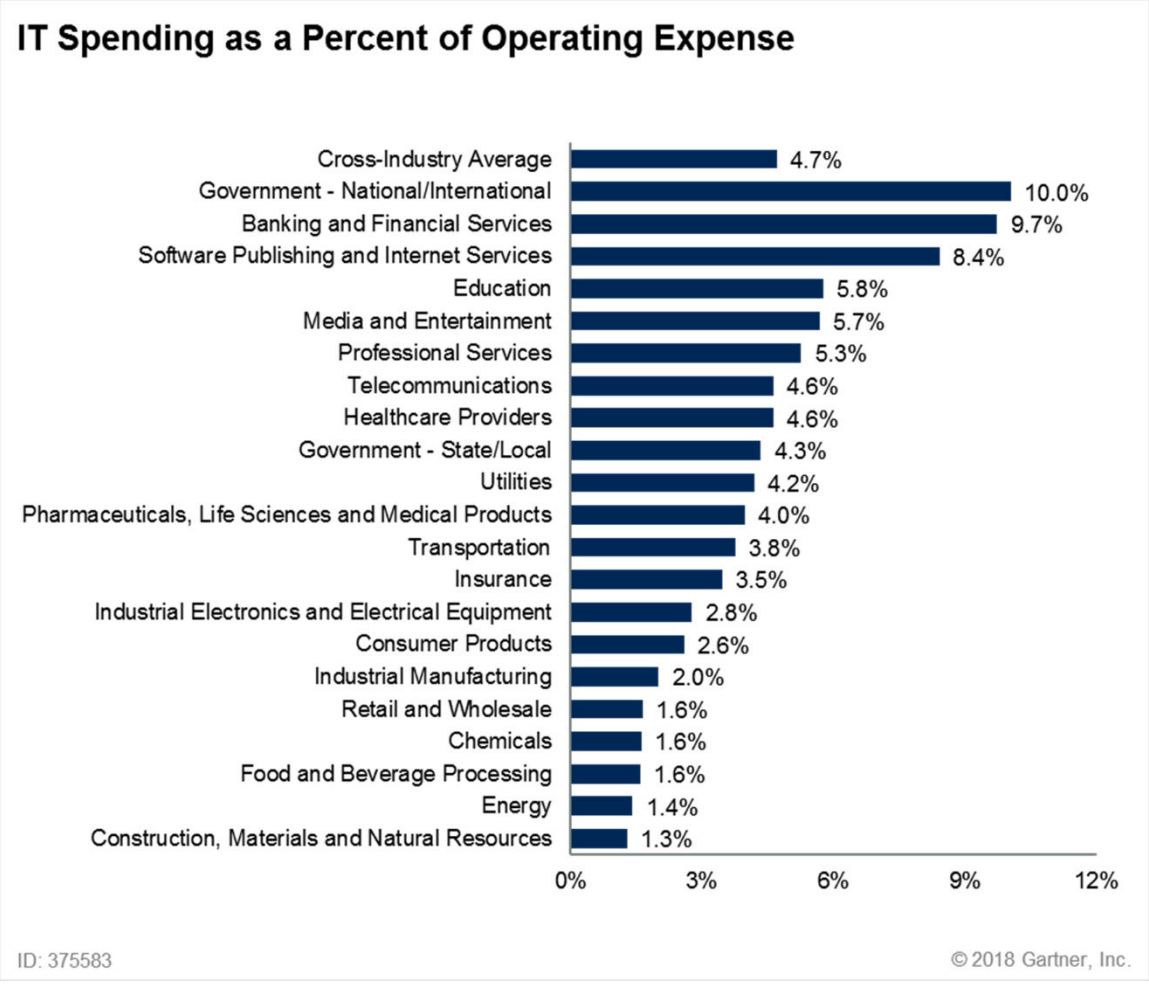
Business operating expense is defined as:

> "The total expense associated with the business units supported by the IT organization. This includes items such as selling, general and administrative expenses, cost of goods sold (or cost of revenue), research and development, depreciation, and depletion and amortization expenses. For insurance, this includes underwriting expenses and loss and loss-adjustment expenses; for banking organizations, it includes interest expenses and noninterest expenses; for government and nonprofit organizations, it is represented by the enterprise operating budget."

While revenue may be subject to external-market-based volatilities, business operating expense typically remains much more consistent and predictable year over year. Therefore, it better reflects the overall business investment strategy. Typically, organizations with a greater level of IT investment relative to operating expense view IT as a strategic enabler, and this can improve business performance and productivity levels.

Industries with lower profit margins will tend to have smaller differences between their IT spending as a percent of revenue and their IT spending as a percent of operating expenses. For example Banking and Financial Services has a profitability of 26.0% while Retail and Wholesale is at 6.2%. IT spending as a percent of revenue and operating expenses for Banking and Financial Services were 7.3% and 9.7% respectively (a difference of 2.4 percentage points). The same numbers for Retail and Wholesale were 1.6% and 1.6% respectively (a negligible difference of less than 0.05 percentage points).

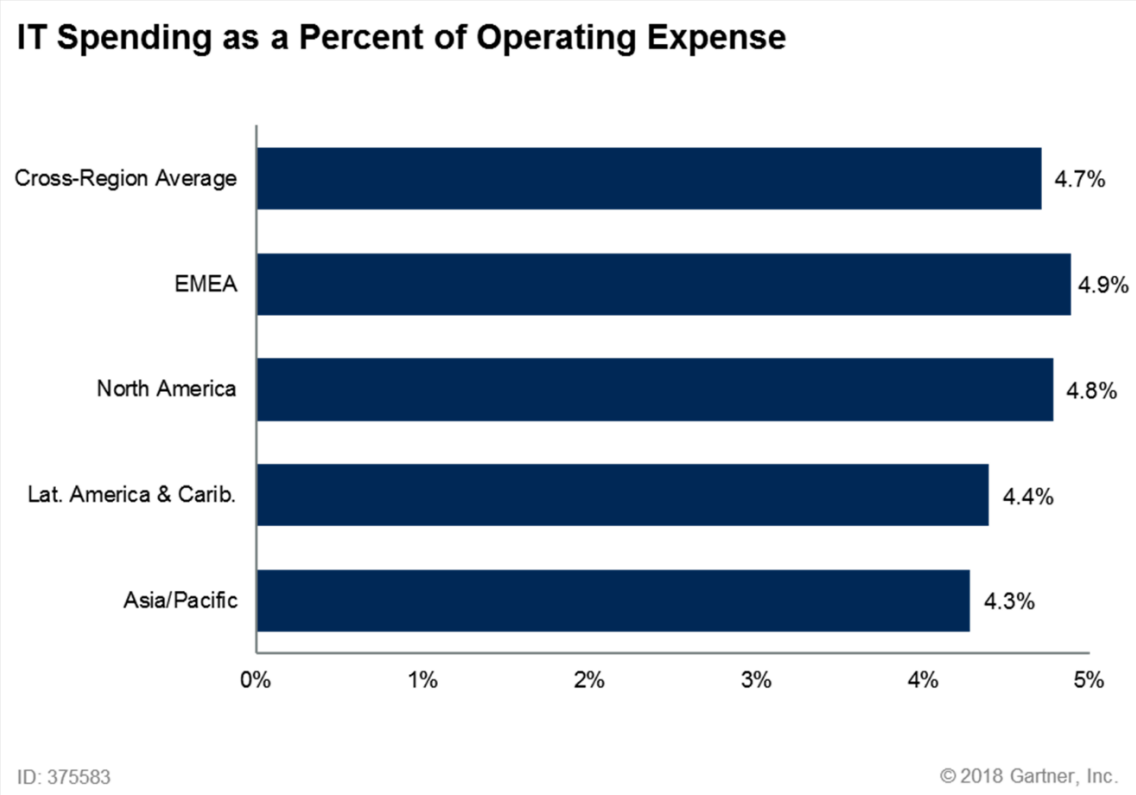
Figure 5. IT Spending as a Percent of Operating Expense, by Industry, 2018



Source: Gartner IT Key Metrics Data (December 2018)

EMEA and North America have the highest levels of average IT of spending as a percent of operating expenses. A key driver behind this difference is the relative cost of staff. The same situation that drove differences in IT spending as a percent of revenue among regions apply here.

Figure 6. IT Spending as a Percent of Operating Expense, by Region, 2018



Source: Gartner IT Key Metrics Data (December 2018)

Again, this metric is calculated on the basis of the current year's IT spending, divided by the previous operating expense. We advise clients to keep this in mind when comparing their own data with Gartner metrics.

IT Spending per Employee, 2018

IT spending per employee is often used to determine the amount of IT support the average organization's workforce receives.

Company employee count is defined as:

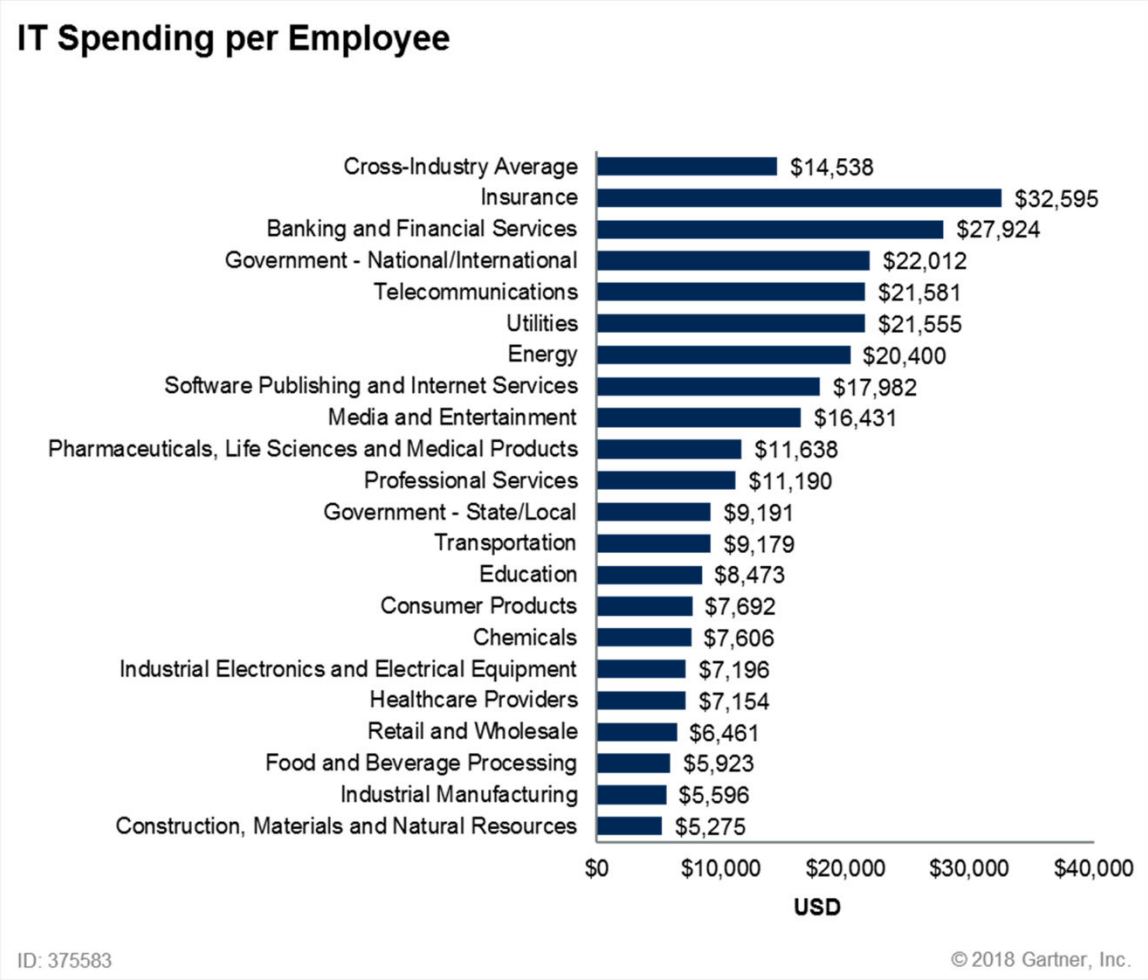
> "The count of employees (i.e., head count, excluding enterprise contractors and consultants) on a full time equivalent basis, regardless of whether these employees are frequent users of the technology supported by the IT organization. This includes full-time and part-time employees, or as reported in the public record."

This measure helps to establish a link between IT investment and automation levels within the context of the workforce that supports revenue. Variations in this measure can represent niche-industry-specific delivery processes for service or product delivery, and, thus, should be viewed in conjunction with revenue and operating income per employee. Organizational staffing strategies and the use of contract employees can also impact this measure.

An increase in IT spending per employee is often viewed as a negative trend. However, this may not always be the case, as a decrease in employees (or a lack of increase of additional employees when business improves) can result in a higher value, simply because there are a smaller number of employees that are divided into the same or increasing IT spending size. Therefore, the trend of increased IT spending per employee still observed in some sectors may have been impacted by other factors, such as a move to digitization driving spending on technology versus personnel, or continuing economic uncertainty in some parts of the world.

The industries with the highest level of IT spending per employee are those that typically tend to be the most information-intensive, and include insurance, banking and financial services, and national and international government (see Figure 7). Industries that are more labor-based, such as construction, materials and natural resources, tend to have much lower IT spending per employee.

Figure 7. IT Spending per Employee, by Industry, 2018

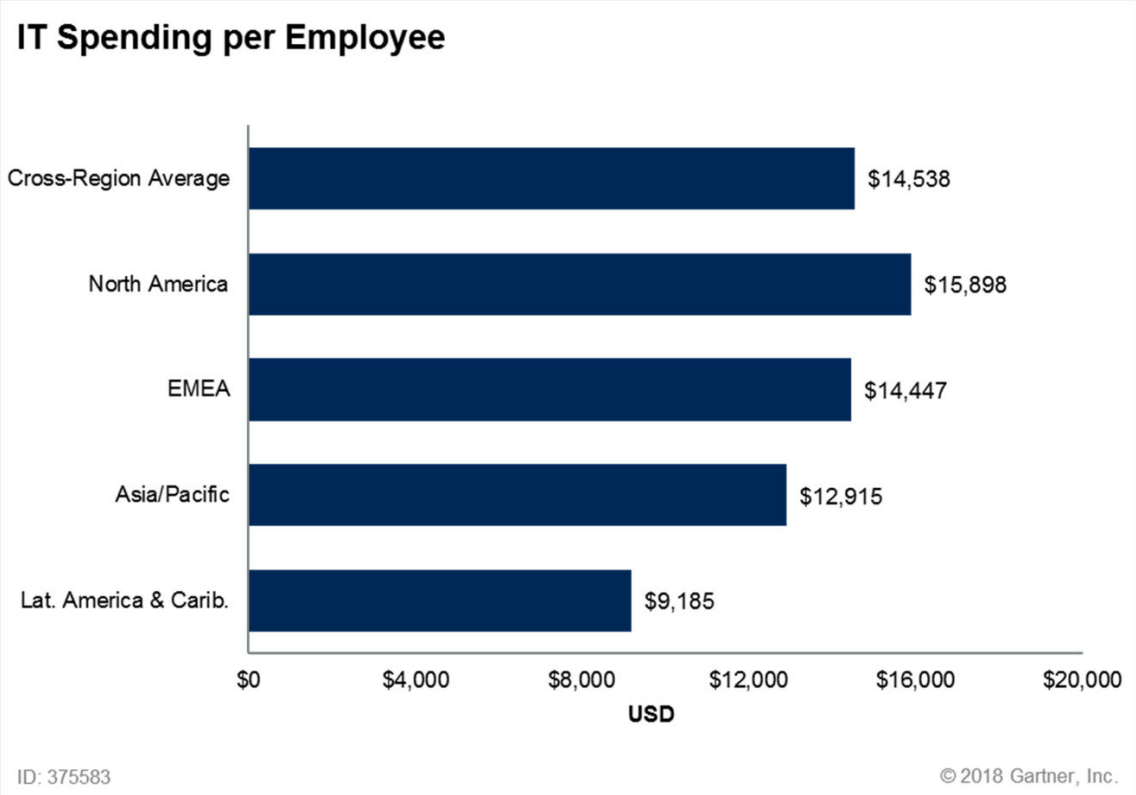


Source: Gartner IT Key Metrics Data (December 2018)

North America has the highest average IT spending per employee, followed by EMEA.

As with IT spending as a percent of revenue and operating expenses, the level of IT spending is lower in regions like Latin America because lower labor costs mean there is less of an incentive to automate. This drives down the IT Spending per Employee. For this metric the fact that there are more employees (the denominator in the metric) in areas with lower staff costs makes the differences even more pronounced.

Figure 8. IT Spending per Employee, by Region, 2018



Source: Gartner IT Key Metrics Data (December 2018)

IT Spending Percent Change, 2017 to 2018

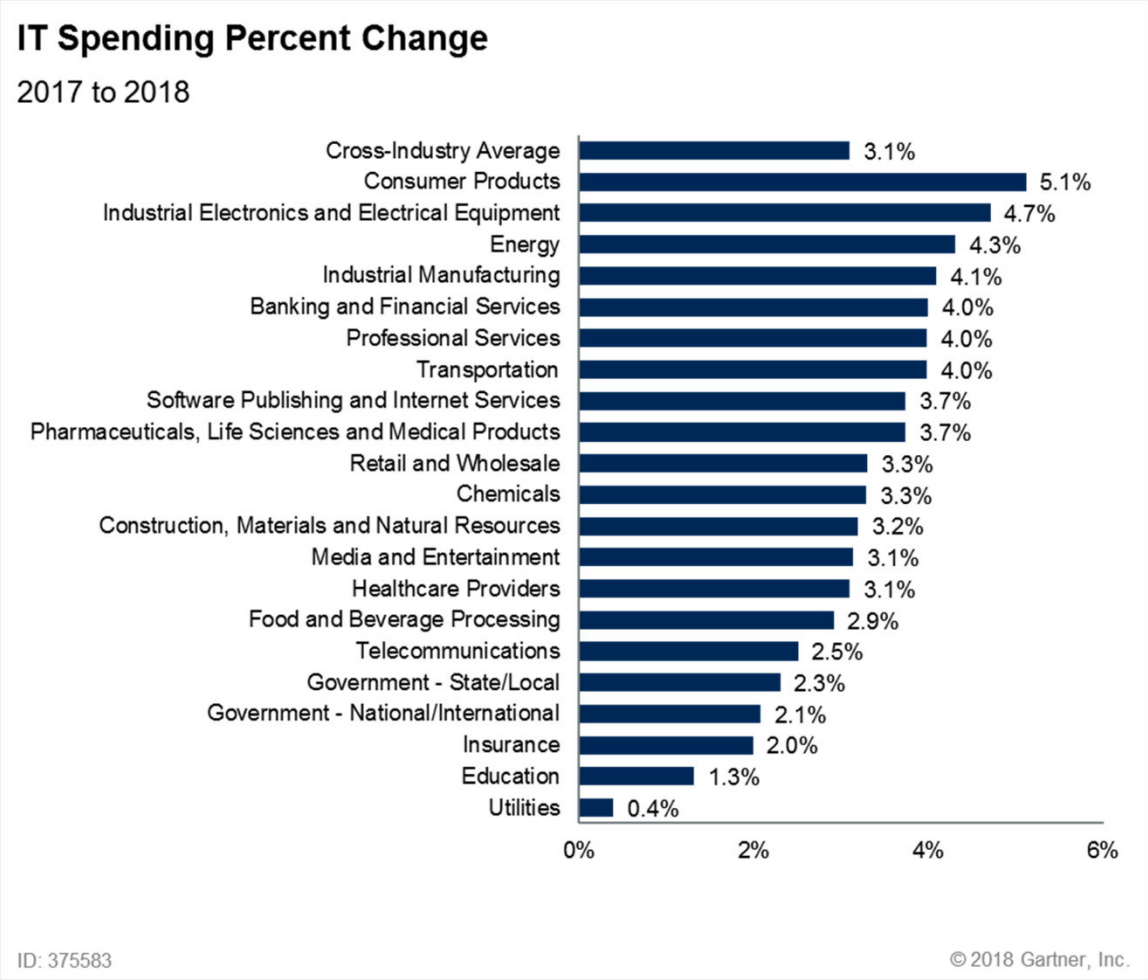
IT spending percent change helps to put context around the directional movement of annual IT spending/budget levels.

There is high visibility around total IT investment figures as they relate to the business. Therefore, understanding the directional movement of the numerator in these ratios is equally as important, because IT investment enables growth and transformation.

The majority of industry sectors continued to see an increase in IT spending/budgets in 2018, with an overall average increase of 3.1% (see Figure 9). Consumer Products has the highest average increase at 5.1%, while education and utilities saw the lowest at 1.3% and 0.4% respectively.

Although average IT spending/budget have been increasing, this does not mean that all organizations saw the same results. The reasons for change in IT spending patterns in organizations can be varied, and can include wider economic and political issues, technology disruptions, business strategy, and investment lifecycles.

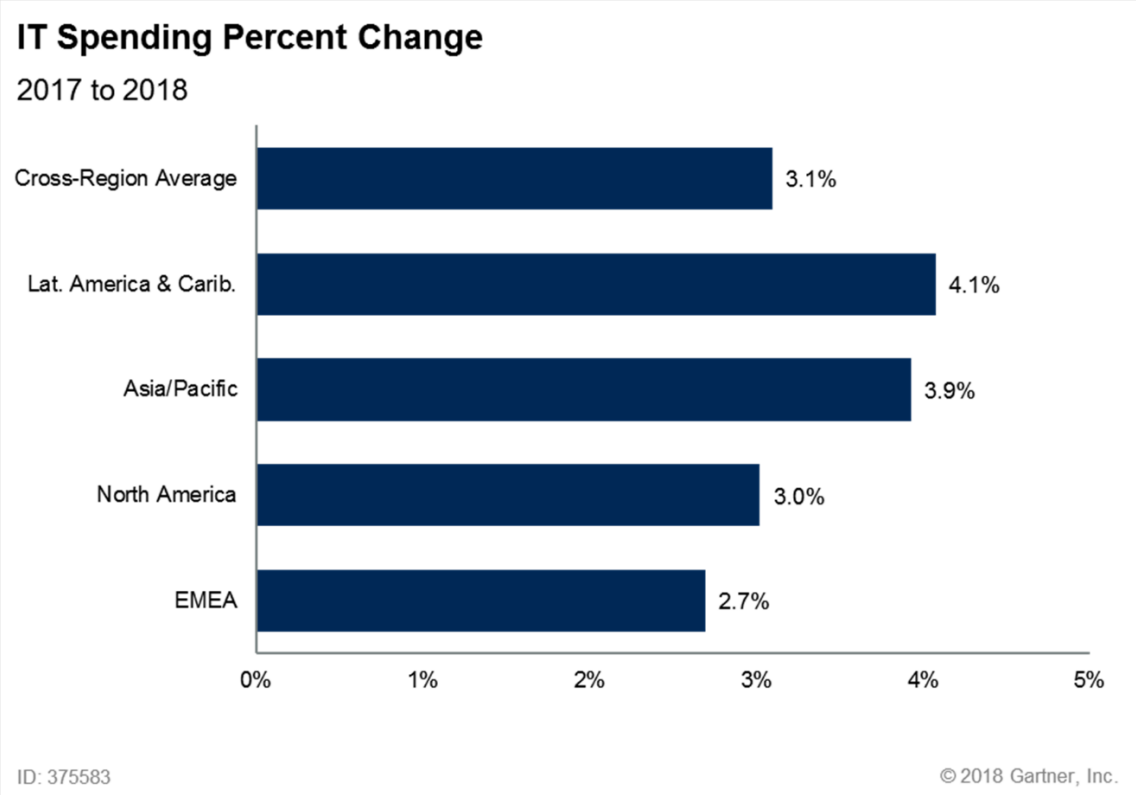
Figure 9. IT Spending Percent Change, by Industry, 2017 to 2018



Source: Gartner IT Key Metrics Data (December 2018)

Regionally Latin America showed the highest increase in IT spending/budgets between 2017 and 2018, at 4.1% (see Figure 10). While EMEA saw a lower level of increase than other regions.

Figure 10. IT Spending Percent Change, by Region, 2017 to 2018



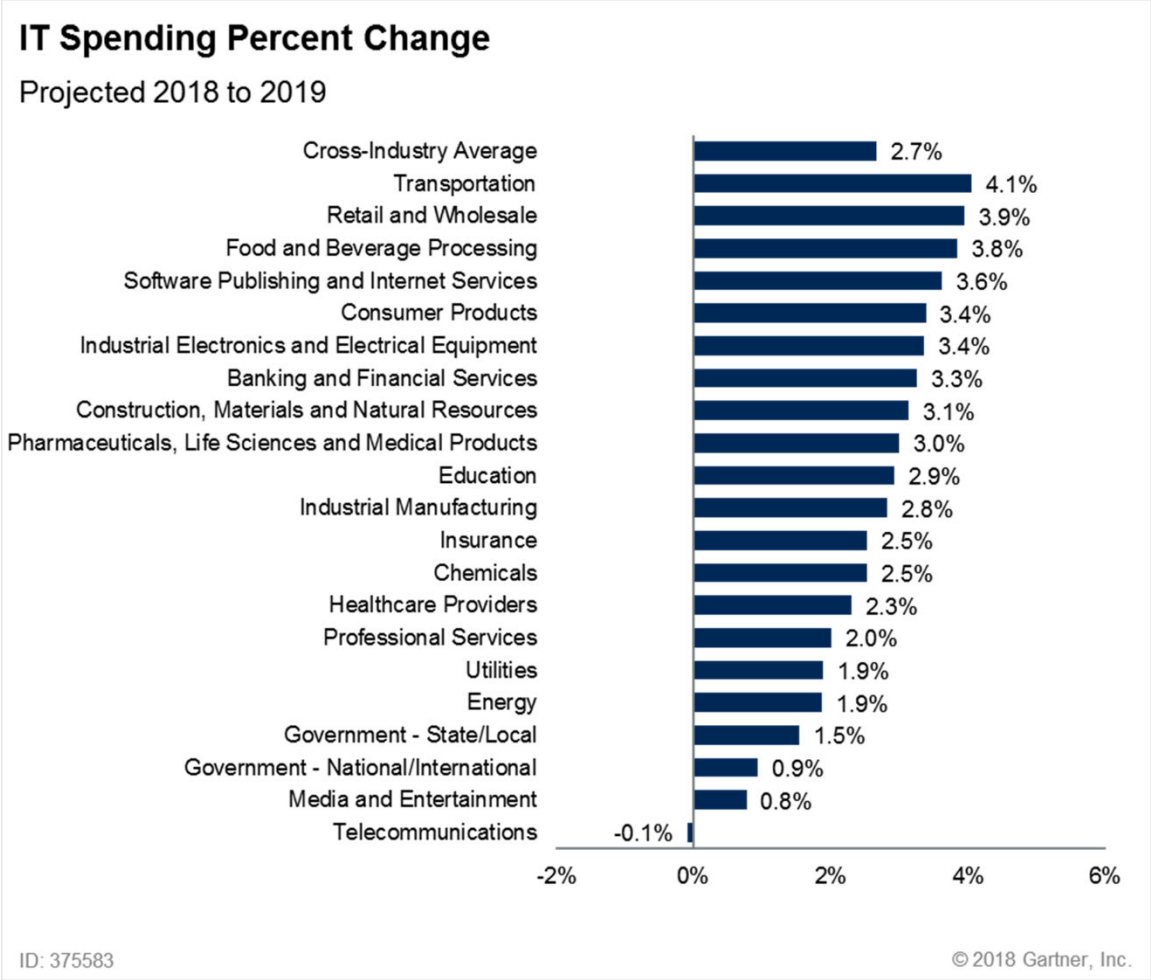
Source: Gartner IT Key Metrics Data (December 2018)

2019 Outlook

IT Spending Percent Change, 2018 to 2019

The outlook for 2019, based on pre-year budgeting information provided by Gartner clients, indicates an overall average planned increase in IT spending/budgets of around 2.7%. And on average all industries are projecting an increase in IT spending/budgets in 2019, except Telecommunications. Figure 11 shows that the level of projected increase varies by sector, and which may be related to industry-specific drivers, such as disruptions, business models and economics.

Figure 11. IT Spending Percent Change, by Industry, 2018 to 2019 Projected

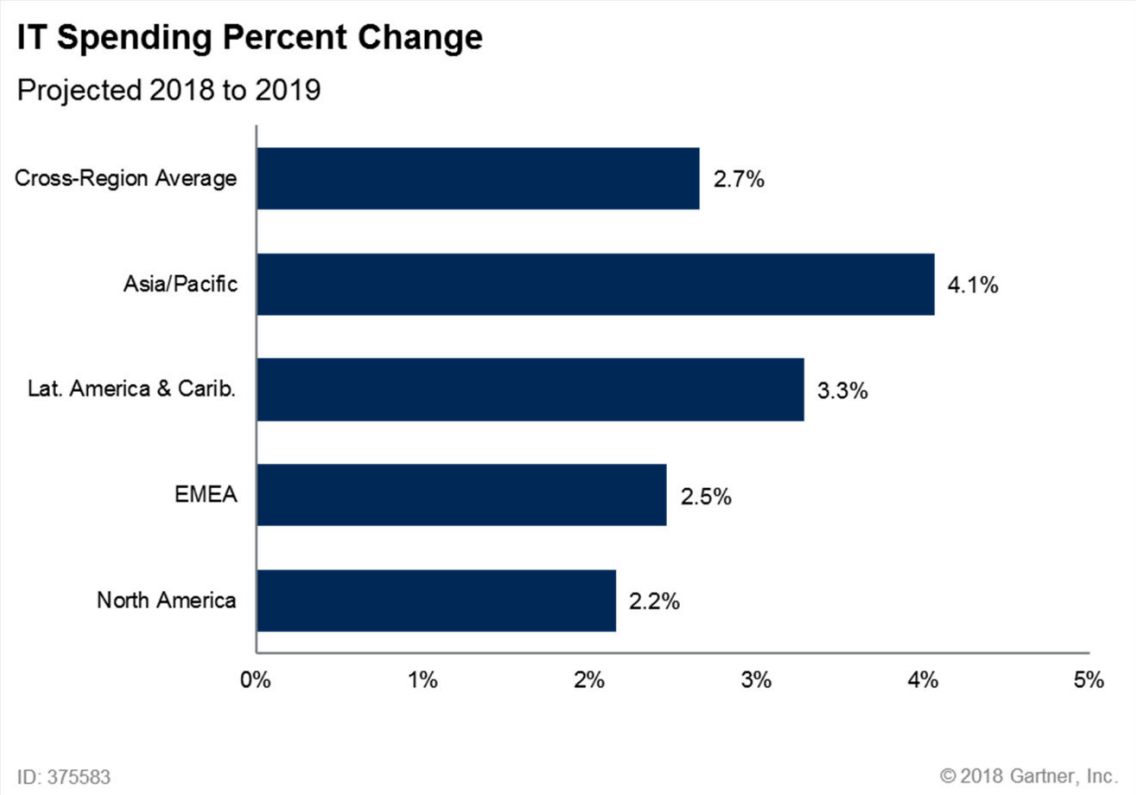


Source: Gartner IT Key Metrics Data (December 2018)

Following the regional trend from last year, Asia/Pacific and Latin America are continuing to project larger levels of increase in IT spending/budgets in 2019, compared with North America and EMEA. (see Figure 12). However Latin America and Asia/Pacific do typically start from a lower base (e.g., lower IT spending per employee), so in real terms the actual amounts may not be as substantial.

The fact that some regions may show a higher average increase than some industry sectors indicates that it is important to consider both industry-specific and local economic conditions when using the ITKMD reports.

Figure 12. IT Spending Percent Change, by Region, 2018 to 2019 Projected



Source: Gartner IT Key Metrics Data (December 2018)

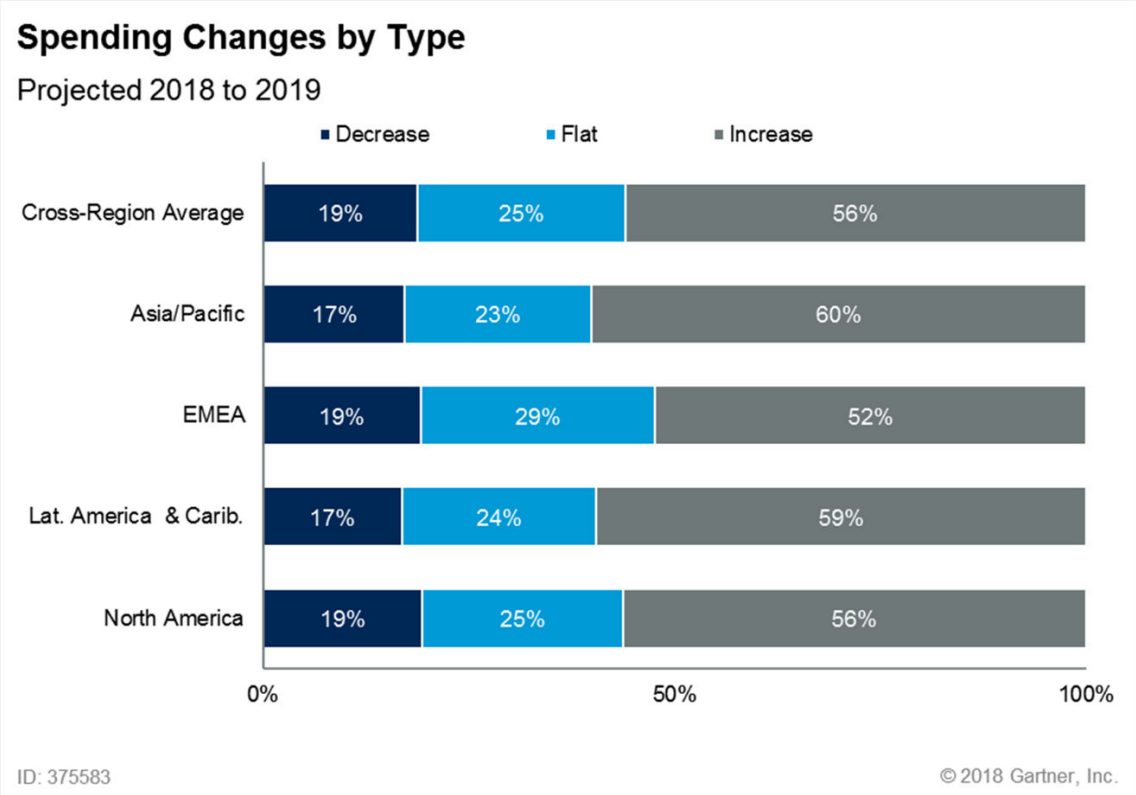
IT Spending Trends: Decreasing Versus Increasing

By looking at the distribution of organizations increasing, decreasing and keeping spending flat, it is possible to get a better understanding of what's driving the averages seen in earlier figures. Also, while some of the overall averages may initially indicate only small increases, to a certain extent this hides the fact that not all organizations are necessarily moving in the same direction, and those that are increasing can be balancing out those that are decreasing.

Continuing the trend from last year, not only does EMEA expect to have the lowest level of average increase in IT spending/budgets in 2019, but more organizations indicated that they expected IT spending to remain at the same level as in 2018 or experience a decrease.

By contrast, in all the other regions more than half the enterprises expected an increase in their IT spending/budgets, and approximately a quarter expected to remain at the same level as in 2018 (see Figure 13).

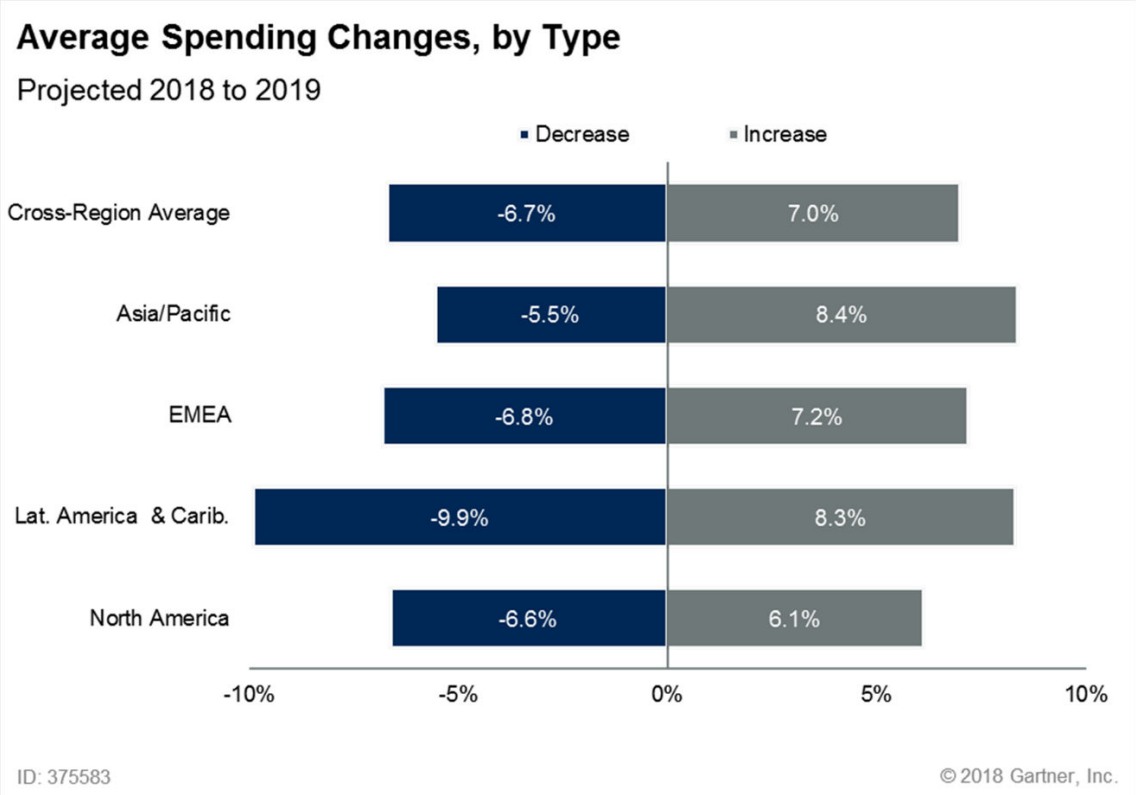
Figure 13. Spending Behaviors: Regional Distribution of Spending Changes by Type, 2018 to 2019 Projected



Source: Gartner IT Key Metrics Data (December 2018)

Figure 14 shows the average IT spending/budget increase and decrease for those that were planning a change (i.e., excludes data points where spending levels were expected to remain the same as in 2018). In Latin America, the average level of decrease is noticeably higher than the other regions. However again these headline averages mask the fact that there are a much wider variety of scenarios in the underlying data.

Figure 14. Spending Behaviors: Regional Average Spending Changes, by Type, 2018 to 2019 Projected

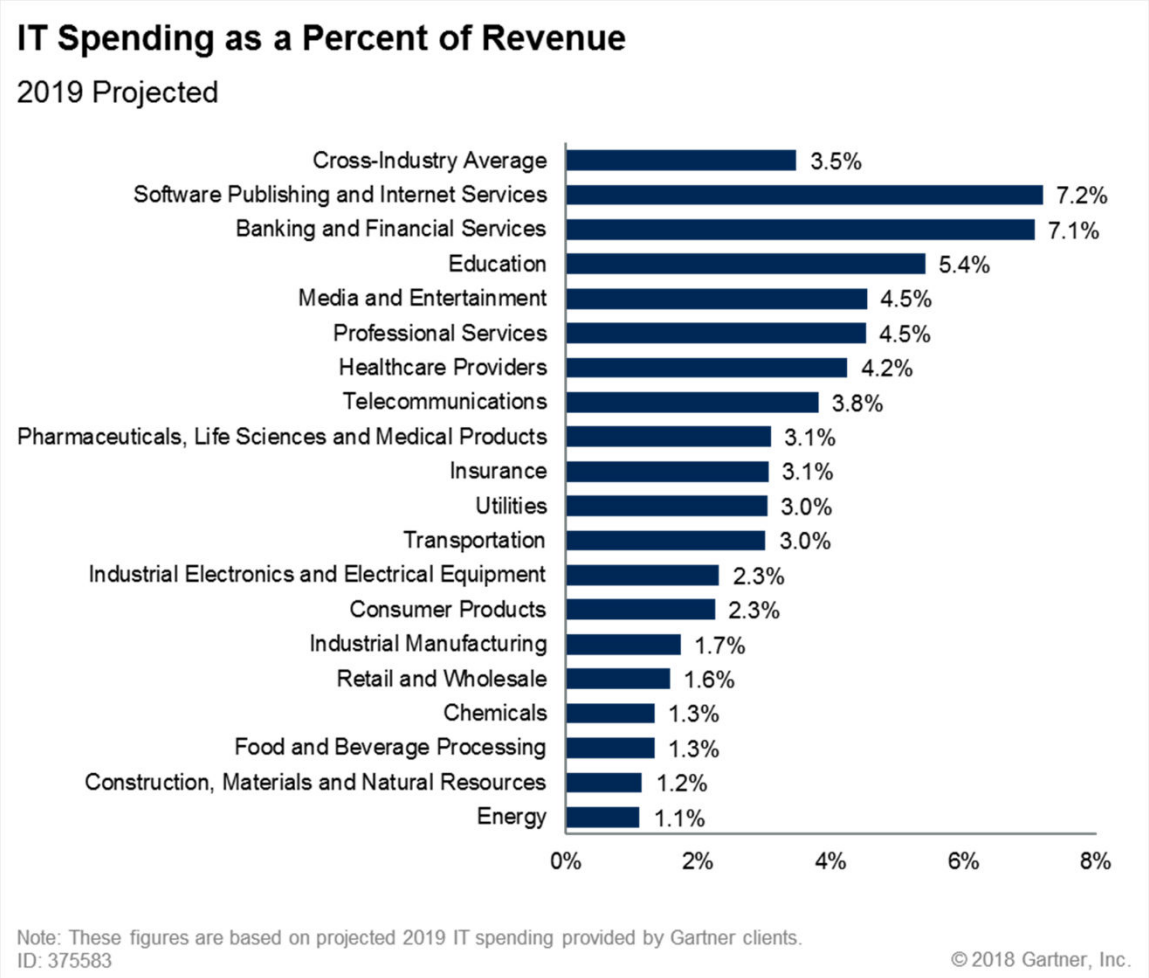


Source: Gartner IT Key Metrics Data (December 2018)

IT Spending as a Percent of Revenue, 2019 Projected

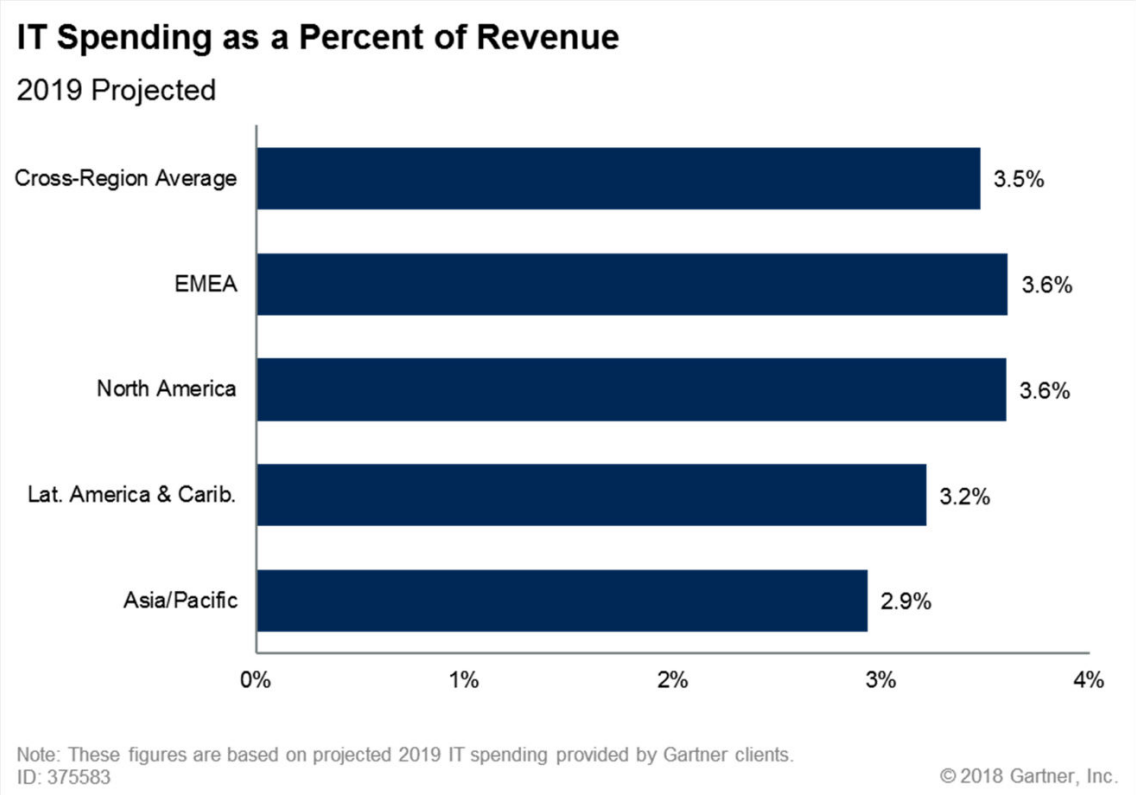
Figure 15 outlines what 2019 IT spending as a percent of revenue is projected to be by industry. This is calculated using the IT Spending change above along with industry level revenue projections from a variety of sources.

Figure 15. IT Spending as a Percent of Revenue, by Industry, 2019 Projected



Source: Gartner IT Key Metrics Data (December 2018)

Figure 16. IT Spending as a Percent of Revenue, by Region, 2019 Projected

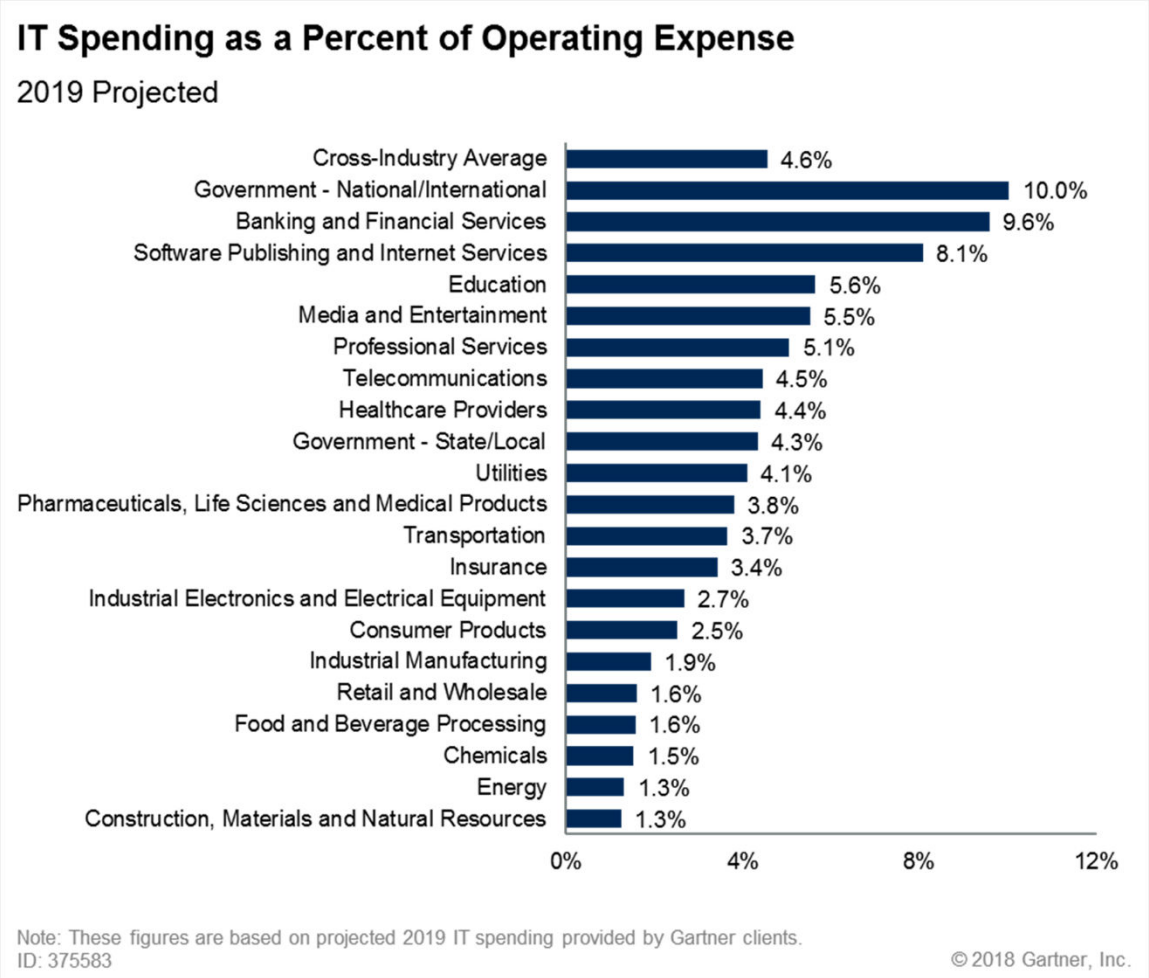


Source: Gartner IT Key Metrics Data (December 2018)

IT Spending as a Percent of Operating Expense, 2019 Projected

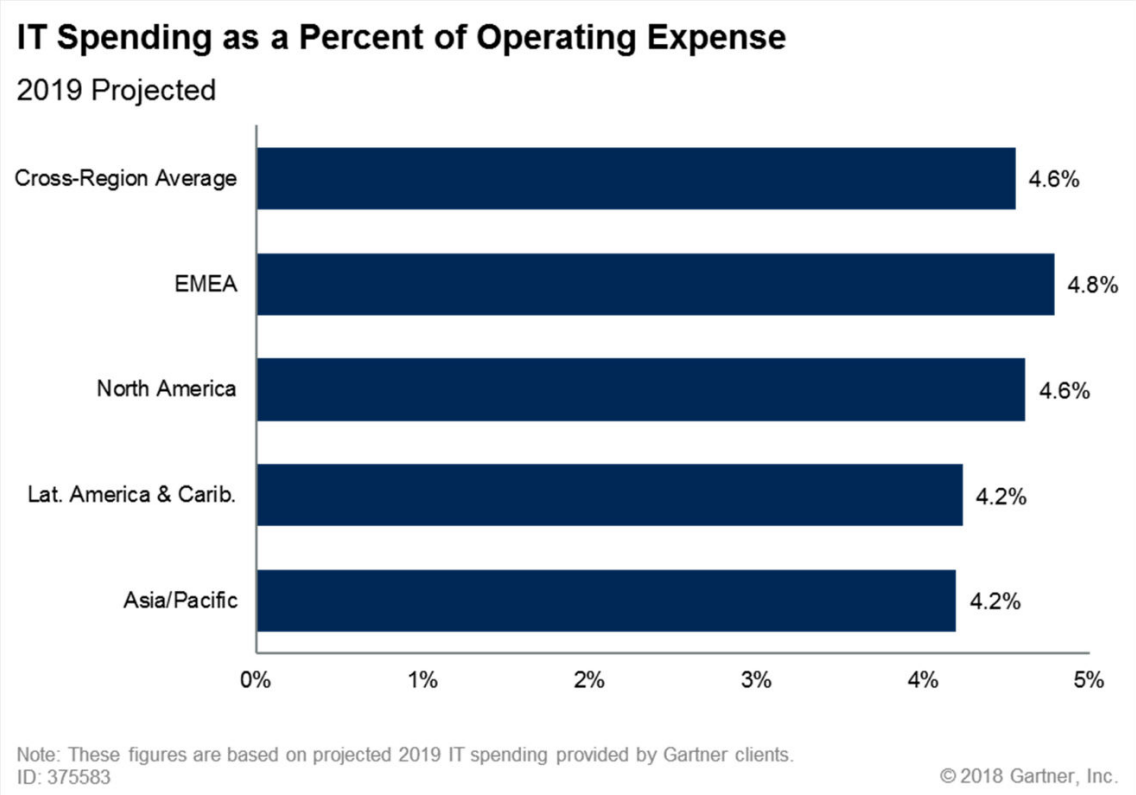
Figure 17 outlines what 2019 IT spending as a percent of operating expense is projected to be by industry. This is calculated using the IT Spending change above along with industry level operating expense projections from a variety of sources.

Figure 17. IT Spending as a Percent of Operating Expense, by Industry, 2019 Projected



Source: Gartner IT Key Metrics Data (December 2018)

Figure 18. IT Spending as a Percent of Operating Expense, by Region, 2019 Projected

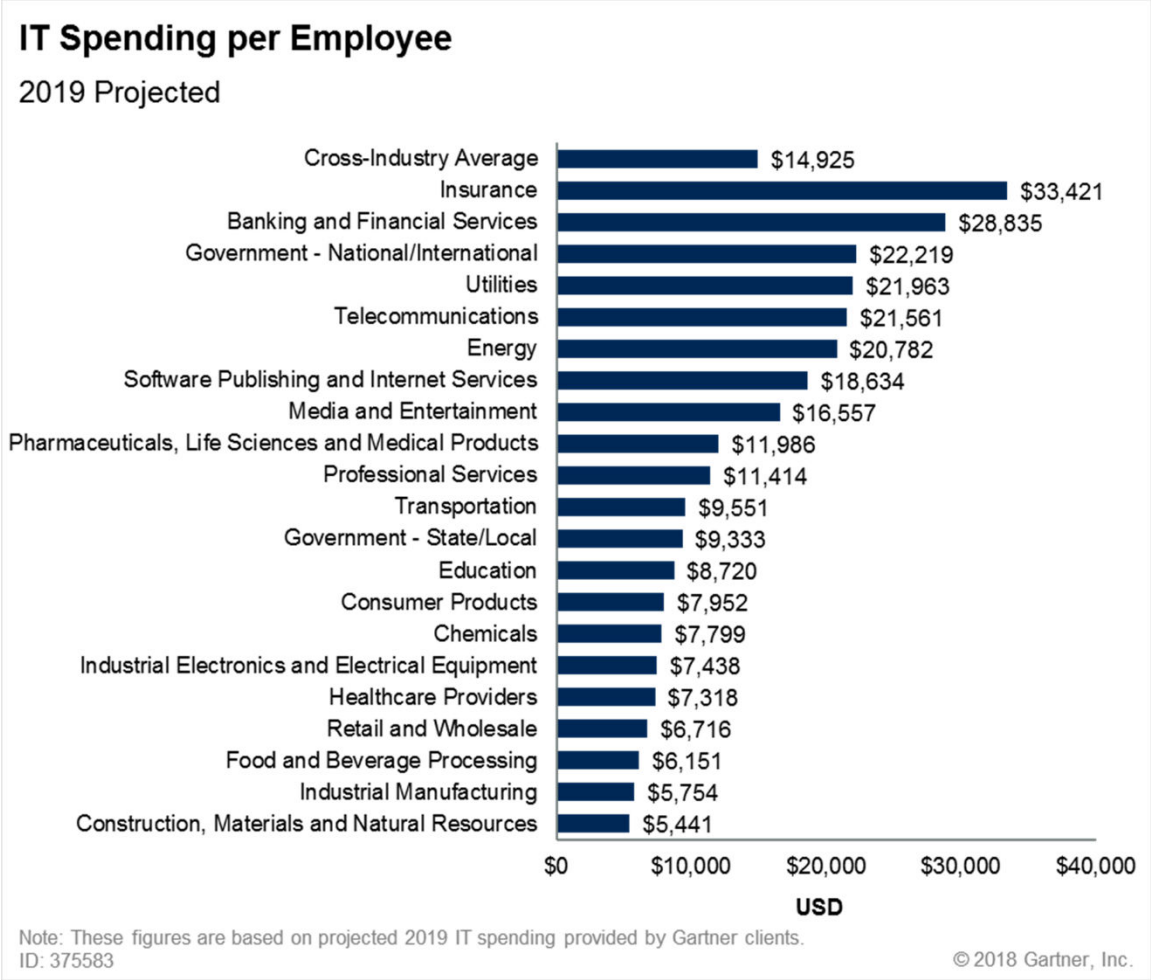


Source: Gartner IT Key Metrics Data (December 2018)

IT Spending per Employee, 2019 Projected

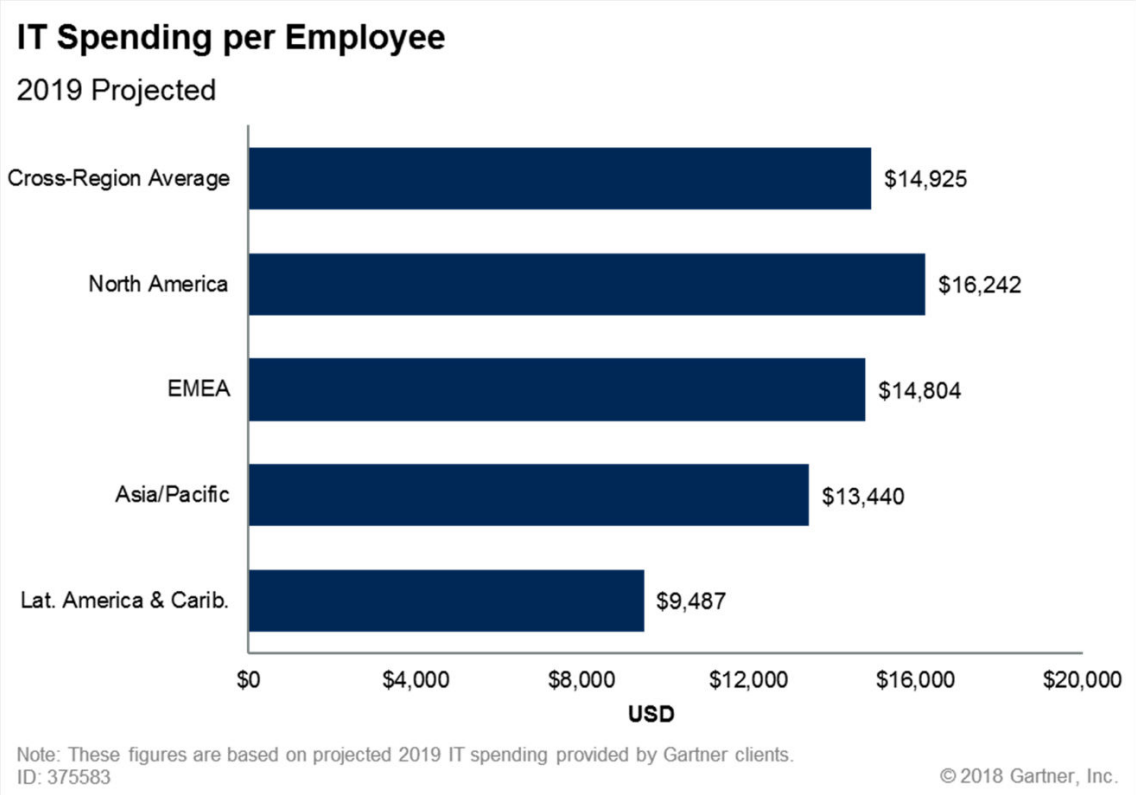
Figure 19 outlines what 2019 IT spending per employee is projected to be by industry. This is calculated using the IT Spending change above.

Figure 19. IT Spending per Employee, by Industry, 2019 Projected



Source: Gartner IT Key Metrics Data (December 2018)

Figure 20. IT Spending per Employee, by Region, 2019 Projected



Source: Gartner IT Key Metrics Data (December 2018)

IT Budget Distributions: Uncover the Facts

Up to this point, the figures have shown spending trends overall, without distinguishing between the strategic, financial or operational categories that compose them. Through these categories, you can draw conclusions about critical investment areas, key investment themes and competitive spending levels.

IT Operational Versus Capital Spending

IT operational versus capital spending helps to portray the IT investment profile for an organization in a given year.

IT operational expense is defined as:

> "The total day-to-day operations and maintenance expenses for this fiscal year that have not been capitalized. These do not include any amortization and depreciation expenses."

IT capital spending is defined as:

> "The total capitalized IT spending for the fiscal year (that is, the full value of capitalized assets acquired in the fiscal year). This includes investments in new application development and IT infrastructure."

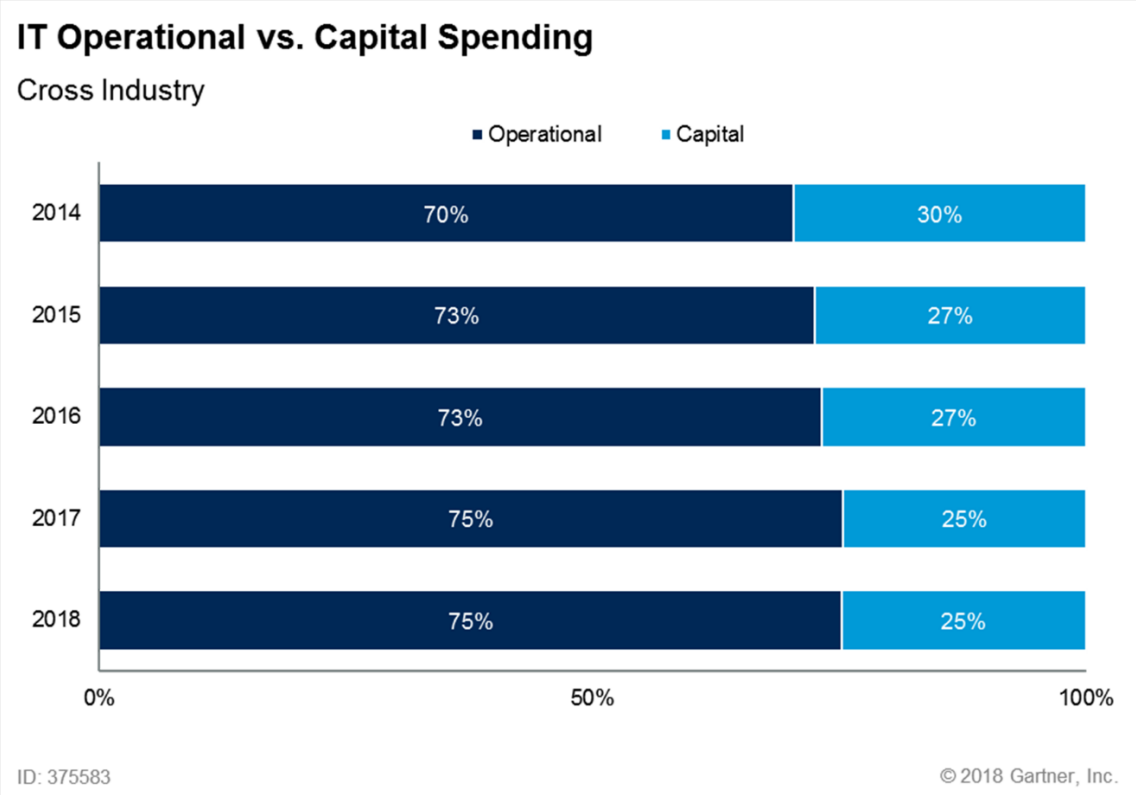
This information is typically available in most accounting or IT finance departments, and, thus, it may be easy to obtain year over year. This metric can provide visibility into the cyclical nature of capital investments (such as hardware, software and large service contracts) compared with recurring operational expenses (such as personnel, facilities and maintenance expenses). The challenge is in leveraging this information to communicate the linkage between IT investment and business results, because it is a traditional accounting view of IT cash flow and does not highlight how IT investment enables improved business performance.

In 2018 the percentage of IT spending classified as capital remained at same level as 2017 (see Figure 21). We have expected the move to cloud to continue to drive this number down, but that wasn't the case for 2017-2018. It is possible that other capital investments have come into play and negated the effect of the transition to cloud on the ratio.

It should be noted that when looking at individual organizations, Gartner observes a wide range of different distributions between IT capital and operational expenses, and so the emphasis should be placed on understanding your organization's own position and how that relates to your IT and business strategies, and not simply on trying to confirm to the Gartner averages.

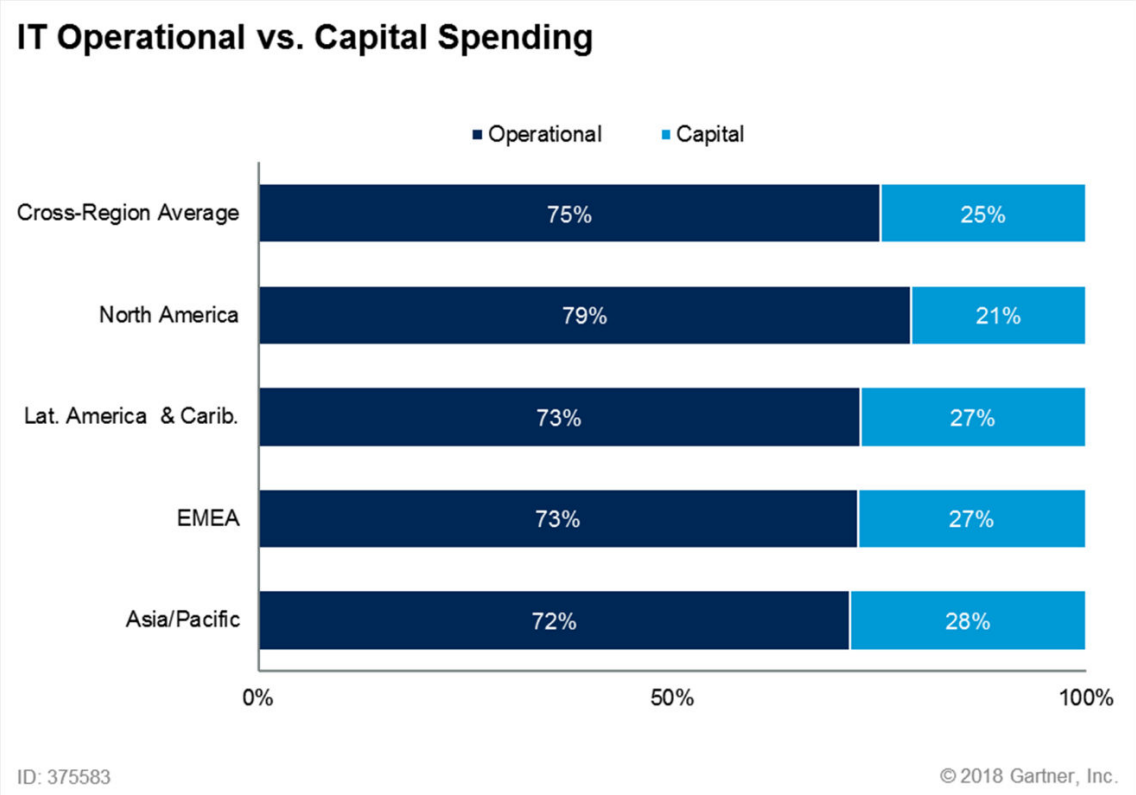
For more information, see "Opex vs. Capex: CIOs Should Partner With CFOs."

Figure 21. IT Operational Versus Capital Spending, 2014 to 2018



Source: Gartner IT Key Metrics Data (December 2018)

Figure 22. IT Operational Versus Capital Spending, by Region, 2018



Source: Gartner IT Key Metrics Data (December 2018)

Strategic IT Spending Portfolio: Run, Grow and Transform the Business IT Spending

The distribution of IT spending to run the business, grow the business and transform the business provides a view of the IT investment profile or "portfolio" to support business performance. In some industries, it is not uncommon to see a high "run" focus — typically because organizations in the industry are not planning strong changes in business model growth or high organic growth — which often translates into a more "cost center" role for IT in the industry or niche sector.

Classifying IT spending into categories that show impact on business outcomes or success can aid alignment and quantify underinvestment in IT. Gartner uses the following portfolio spending categories and defines them as follows:

- **Run the business:** This is an indicator of how much of the IT resource is consumed and focused on the continuing operation of the business. It includes all nondiscretionary expenses as part of the run-the-business cost. Some businesses call this "business as usual," "keep the lights on" IT spending, or sustain investments. Run expenses do not directly increase revenue, or achieve by themselves new or enhanced goals of the enterprise.
- **Grow the business:** This is an indicator of how much of the IT resource is consumed and focused on developing and enhancing IT systems in support of business growth (typically

organic growth). Discretionary investments are more likely to be included in the grow-the-business or transform-the-business cost.

- **Transform the business:** This is an indicator of how much of the IT resource is consumed and focused on implementing technology systems that enable the enterprise to enact new business models. This is very much a "venture" category and would be represented by activities such as an insurer introducing usage-based insurance products such as telematics or a supermarket combining real time analytic monitoring with in-store task management to provide automated alerts to store staff to perform preemptive tasks.
- **Change the business:** This combines the grow and transform the business IT spending categories for the sake of simplicity. If change the business IT spending is used as a category, this often means that grow and transform categories are not used (thus, run versus change the business IT spending). Use of this categorization is often favored where there is ambiguity as to the classification of expenses or investments, but where there is still a clear distinction between run and change the business investments.

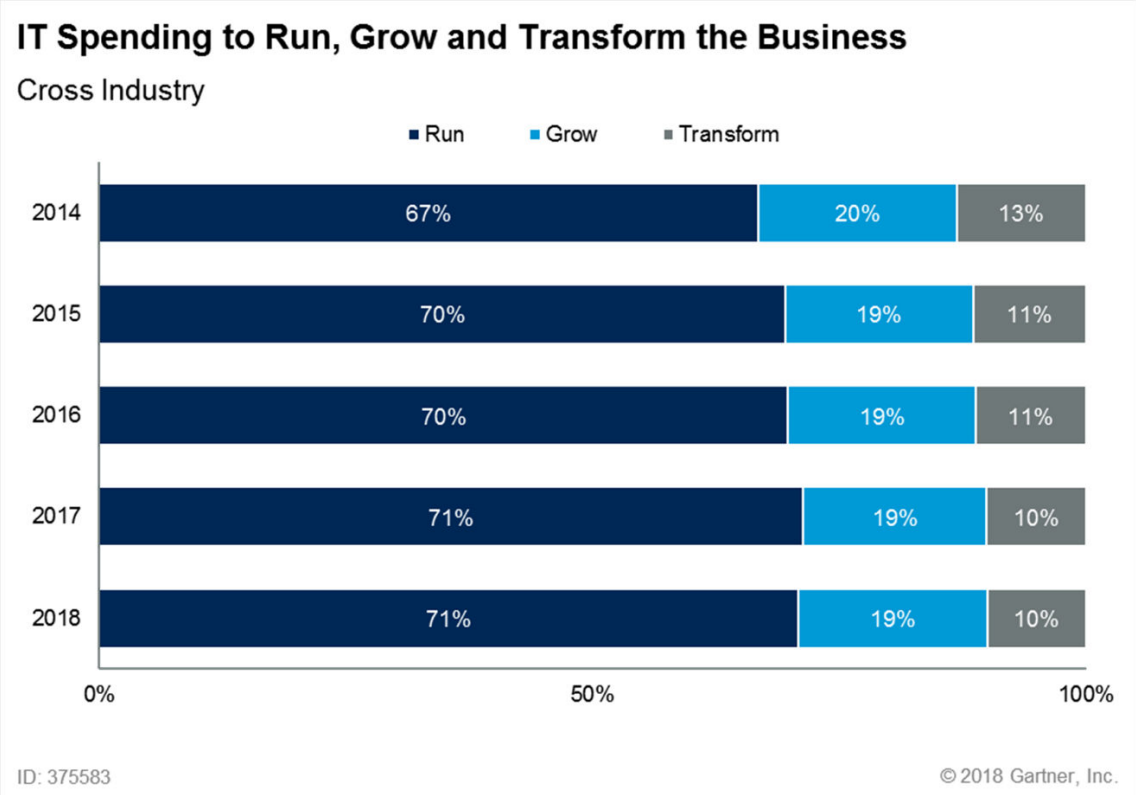
Gaps in business alignment can be found by examining IT spending as it relates to the day-to-day operations of a business (run), the organic growth of the business or productivity improvement (grow) and its support with new revenue creation from major business transformation, new products, services or business models (transform).

A common misconception with this measure is that an IT initiative that may transform the IT organization, such as data center modernization or virtualization, should be classified as a "transform the business" investment. While these IT initiatives do transform the IT organization, they should primarily be classified as "run the business" investments because they support pre-existing IT services. IT transformation often leads to new business process improvements that enable the business to grow or build new revenue streams. Therefore, these costs would need to be evaluated and distributed based on IT service and business performance.

The run, grow and transform business framework should always be viewed in business terms with respect to how IT will enable the business to grow or transform revenue, operating income and/or profit margins.

The distribution of IT spending among run, grow, and transform may shift slightly from year to year. The fact that anything introduced as a grow or transform activity in one year will eventually become a run activity in the future tends to keep this data relatively stable.

Figure 23. IT Spending to Run, Grow and Transform the Business, 2014 to 2018



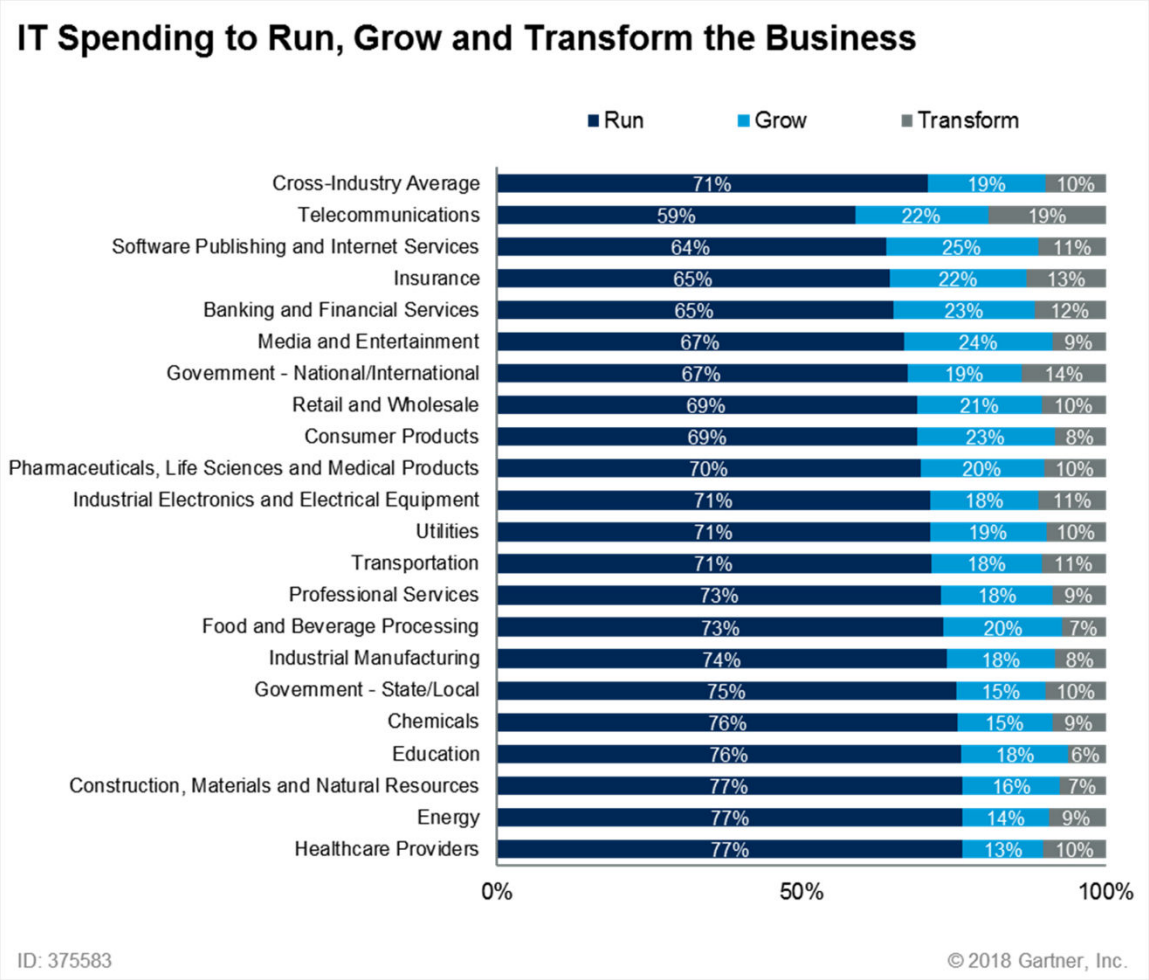
Source: Gartner IT Key Metrics Data (December 2018)

Subtle hints of an industry's IT investment profile can be seen in a higher percentage of IT spending devoted to the grow and transform categories, which comes at the expense of the run category.

Telecommunications, and software publishing and internet services, have the highest percentages of change the business, while energy and healthcare providers has the lowest (see Figure 24).

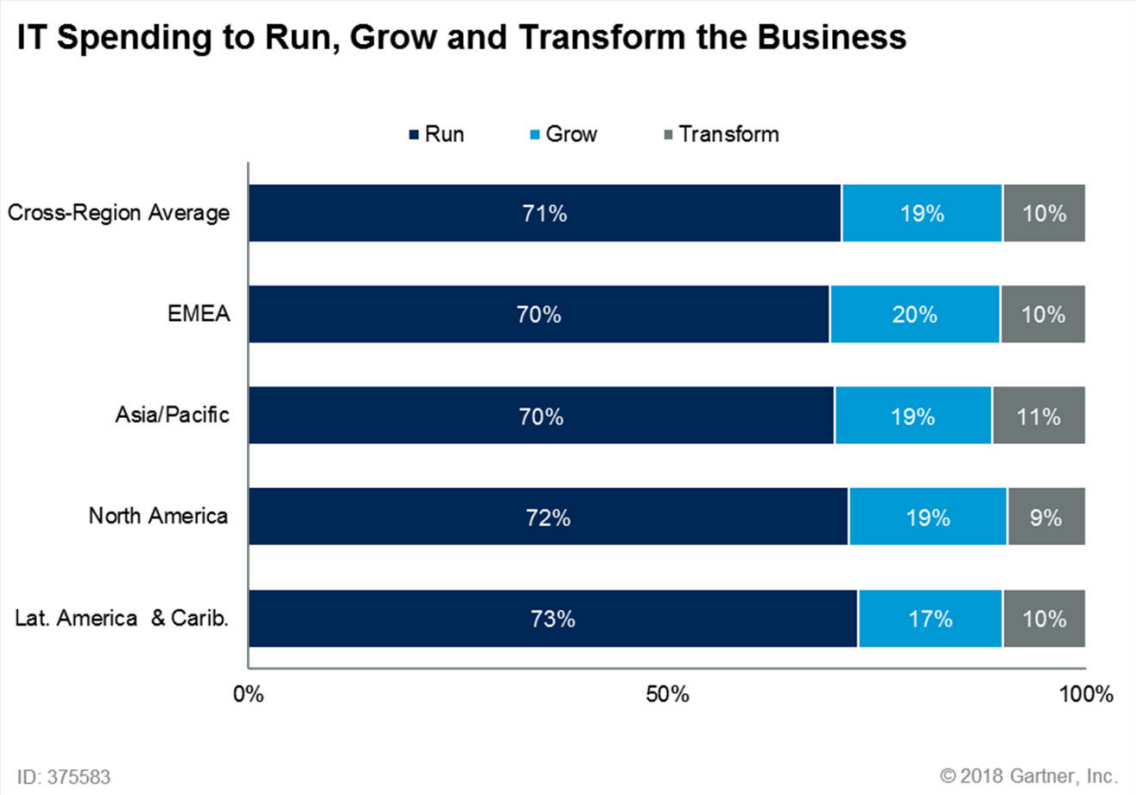
Just because an industry has a higher percentage of “run” spending doesn’t mean that there isn’t a good deal of growth and innovation happening. For example, an industry like education has an average percent spending on run of 76% which is higher than 69% for retail and wholesale. It is important to remember that education has an average IT spending as a percent of revenue of 5.6% versus 1.6% for retail and wholesale. Because education is starting out at a higher level of IT spending it may be investing more in IT to grow and innovate than retail and wholesale is.

Figure 24. IT Spending to Run, Grow and Transform the Business, by Industry, 2018



Source: Gartner IT Key Metrics Data (December 2018)

Figure 25. IT Spending to Run, Grow and Transform the Business, by Region, 2018



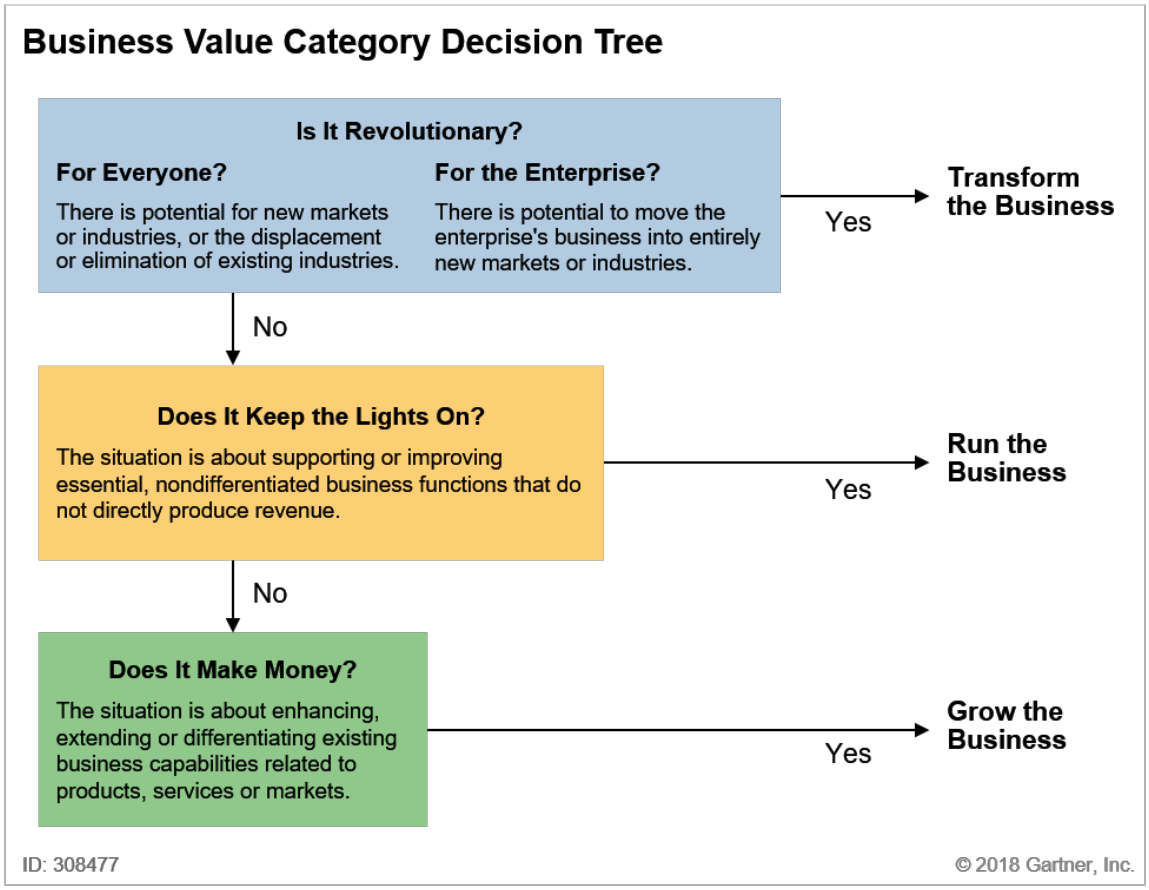
Source: Gartner IT Key Metrics Data (December 2018)

Determining the Business Context for Value

As organizations leverage the run, grow and transform the business concepts at a macro level, Gartner has found it helpful to define various IT investments (and portions of investments) with the same basic framework to illustrate the projected impact at the individual IT initiative and project levels.

With a basic understanding of the framework, as outlined here, organizations can apply the decision tree (see Figure 26) to select the category that best describes business value for their IT initiatives.

Figure 26. Business Value Category Decision Tree



Source: Gartner (December 2018)

The Link to Strategy

The run, grow and transform the business framework is a starting point for the overall process of measuring, forecasting and communicating IT value. Gartner believes that the initial language and metrics used for business value are critical success factors in the organization's ability to make good IT investment decisions. For organizations that are looking for best practice, consider linking individual IT services to individual business process performances in a causal chain.

For more information on run, grow and transform the business, see "Run, Grow and Transform the Business IT Spending: Approaches to Categorization and Interpretation" and "Simplify Service Portfolio Prioritization and Resource Planning Using Run-Grow-Transform Categorization."

IT Spending by Funding Source

IT spending can come from a number of different sources within an enterprise or organization, and is not restricted to the formal IT Budget. Additional spending can occur within business unit budgets and be what is known as "shadow IT." The sources of IT spending are defined as follows:

- **Formal IT Budget**: IT spending that the IT department is accountable for. The IT department is answerable for the provision of these IT assets and services.
- **Business Unit IT**: IT spending where profit centers or overhead departments are accountable. For this category the IT department is consulted as a subject matter expert, and there is two-way communication.
- **Shadow IT**: IT spending anywhere in the enterprise for which the IT department is not accountable. For this category the IT department is aware of spending, but may not have detailed information about it. Amounts here may need to be estimated.

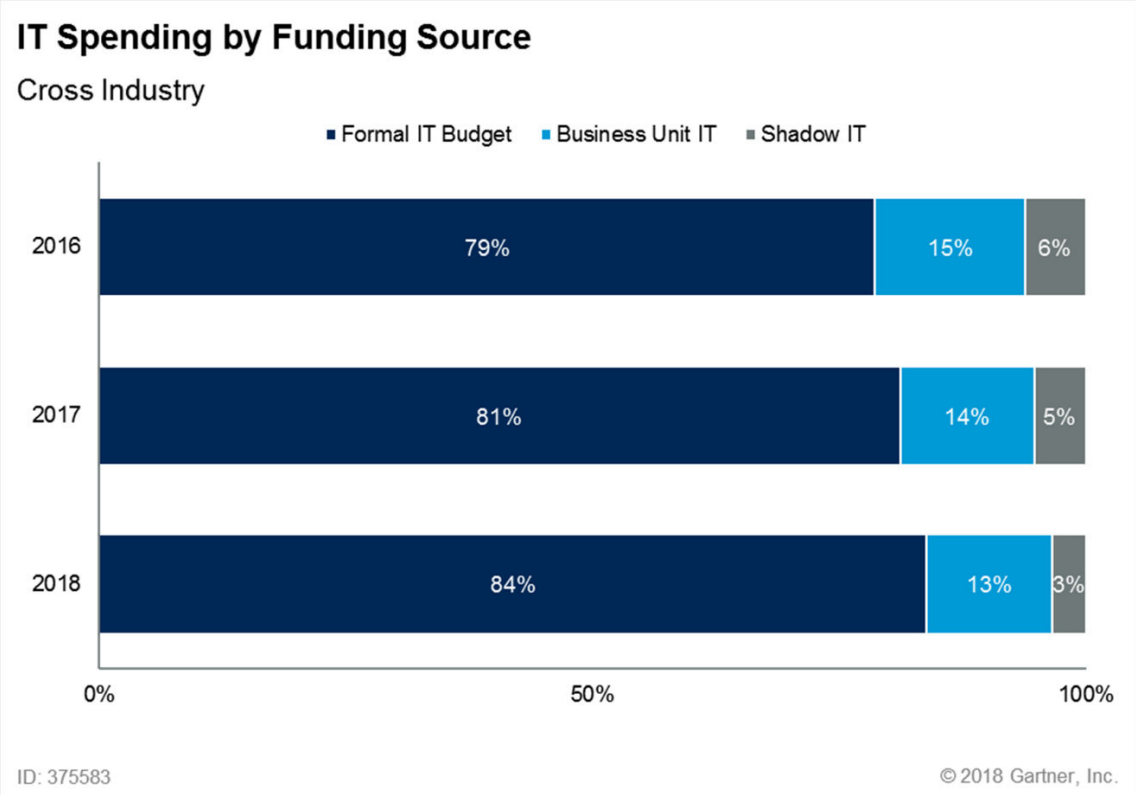
For the purposes of this metric the IT department is defined as the formal organization headed by the CIO/senior IT leader of the entity being analyzed. It also includes any IT organization reporting into them.

Understanding how much IT spending occurs outside the formal IT budget allows organizations to gauge the true extent of their IT spending, and ensure for example that IT budget cost cutting exercises do not simply result in IT spending occurring elsewhere in the business. Getting the right mix of the formal IT budget, business unit IT and shadow IT can depend upon many factors, and needs to be appropriate for the circumstances of the individual organization. Shadow IT can occur because the business wants to move faster than the formal IT departments processes allow, and can lead to a lack of central governance and control. However not all shadow IT should necessarily be view as "bad."

See "Integrate ‘Shadow IT’ and Business-Led IT Into the I&T Operating Model to Enable Enterprise Agility" for more details.

The data over the past three years has shown an increase in the percent IT Spending from the formal IT budget from 79% to 84%. This seems counterintuitive as anecdotally we hear that business units are taking more responsibility for IT Spending with the move to digital business models. This may be true to some extent, but when we dig into what people call “Shadow IT” we often find that much of it includes assets and services that aren’t part of the IT Spending/Budget Definition. What people casually call “Shadow IT” often includes things like digital advertising, power users of business intelligence applications, BPO, and content functions.

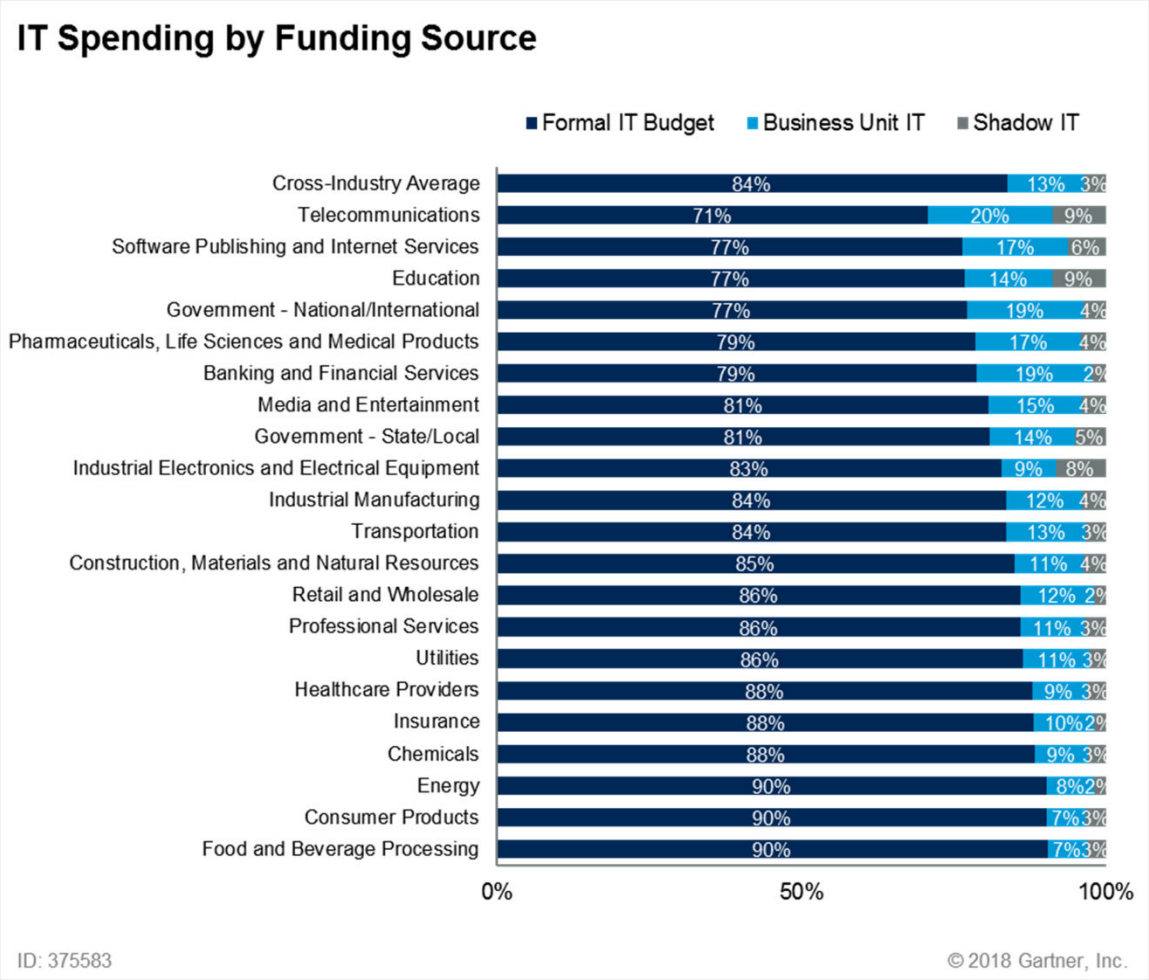
Figure 27. Distribution of IT Spending by Funding Source, 2016 to 2018



Source: Gartner IT Key Metrics Data (December 2018)

Certain industries have a more of a culture of decentralization of management of IT resources and that is evident in Figure 28. In higher education, it is not uncommon for academic departments to have a high degree of control over their IT spending. We see that they are the third lowest in percentage of IT Spending from the formal IT budget. Some of the industries with higher levels of percentage of IT spending from the formal IT budget are the more conservative, more centrally managed ones like insurance and energy.

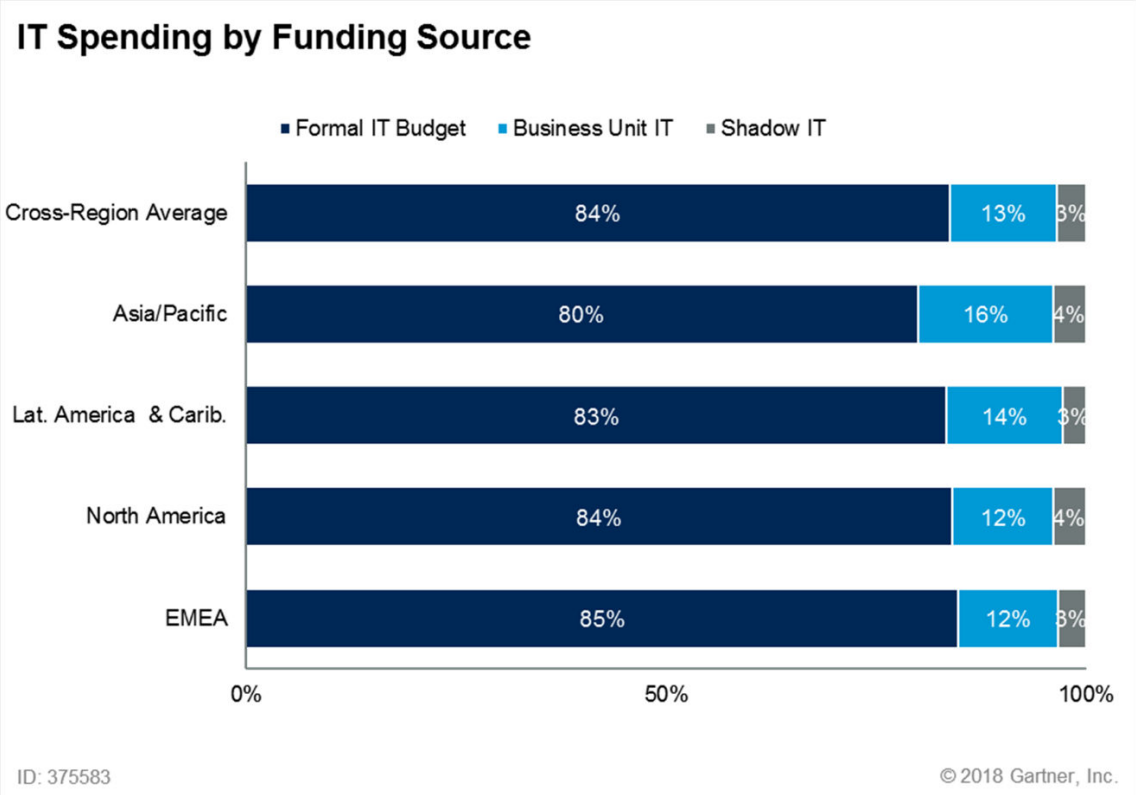
Figure 28. Distribution of IT Spending by Funding Source, by Industry, 2018



Source: Gartner IT Key Metrics Data (December 2018)

Regionally North America and EMEA have the highest percentage of formal IT budget (see Figure 29), while Asia/Pacific has the largest percentage of business unit IT.

Figure 29. Distribution of IT Spending by Funding Source, by Region, 2018



Source: Gartner IT Key Metrics Data (December 2018)

IT Spending by Asset Category, 2018

The distribution of spending between asset categories (hardware, software, personnel, outsourcing and public cloud) can show the dynamics of IT investments. For the purpose of this research, personnel includes occupancy/facilities costs, outsourcing includes network transmission, and public cloud is a combination of SaaS IaaS and other public cloud services.

The definitions of each category are as follows:

- **Hardware Expenses:** These include all hardware expenses described in the IT spending/budget definition.
- **Software Expenses:** These include all software expenses described in the IT spending/budget definition.
- **Personnel Expenses:** These include:
  - **Salary and Benefits Expenses:** Salary (including overtime pay), benefits and "other" employee costs, such as travel and training for all IT FTEs. The "benefit load" should include costs for bonuses, paid holidays, vacations, medical/dental coverage, life and accident insurance, retirement plans, stock plans, disability, Social Security, unemployment

compensation, dependent care, tuition reimbursements and employee assistance programs (for example, physical exams, exercise programs and similar costs).

For contractors and consultants, include all compensation that was paid directly to the individual or agency.

Do not include the spending related to human resource department staff allocations, early retirement incentive bonuses and internal “cross-charges” for corporate overhead such as for the chairperson's salary.

- **Occupancy/Facilities Expenses:** The fully burdened costs for the facilities being used by the staff that supports the enterprise. Some examples include office space, furniture, electricity, maintenance, property taxes, security and office supplies. Occupancy costs for space dedicated to IT functions, such as the data center (including power/heat management and raised floors), are also included.

- **Outsourcing Expenses**: These include:
  - **Outsourcing:** The fees for third-party or outsourcing contracts in which "outsourcing" is defined as any situation in which the full operational responsibility for IT services is completely handed over to an external service provider (for example a service desk contract based on a per user or per contact fee). Third party private cloud implementations involving single tenancy are included in this category.
  - **Network Transmission:** Outsourced Transmission charges for mobile and fixed voice and data telecommunication services.
- **Public Cloud Expenses**: These include
  - **Software as a Service (SaaS)**: SaaS is the application service layer within cloud computing. The application software is owned, delivered and managed remotely by one or more providers. The provider delivers a solution based on a uniform application definition and on a sharing model at one or more layers of the application stack. SaaS is purchased on a pay-for-use basis or as a subscription based on usage metrics.
    - Customers may be able to extend the data model by using configuration tools supplied by the provider, but without altering the source code. This approach is in contrast to the traditional application hosting model in which the provider supports multiple application codes and multiple application versions or a customized data definition for each customer.
    - Purchasing is based on a subscription (for example, a per-user, per-month fee) or use basis (for example, allocating a certain number of transactions for a fixed time period or an actual pay-per-use basis). A perpetual license purchase isn't considered SaaS.
  - **Infrastructure as a Service (IaaS) and Other Public Cloud Services**.
    - **Infrastructure as a service (IaaS)** is a standardized, highly automated offering in which compute resources, complemented by storage and networking capabilities, are owned and hosted by a service provider and offered to the customer on demand. The

resources are multitenant, metered, and operate in near-real time. Self-service interfaces are exposed directly to the customer, such as a web-based graphical user interface and API.

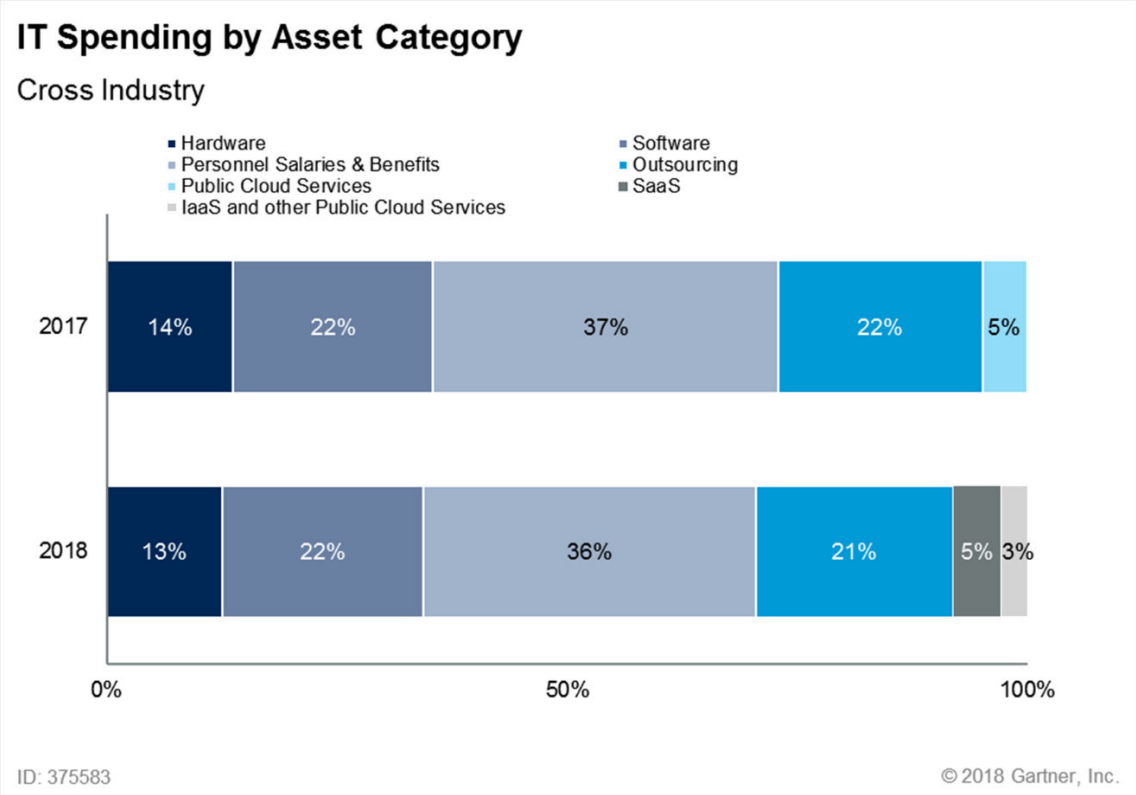
- **Other Public Cloud Services** comprise public cloud management and security services, and platform as a service (PaaS). These services managed by a provider, offered by subscription and in a multitenant manner, with some sharing of resources between tenants to increase overall efficiency and scalability of the operation.
  - **Cloud Management and security services** include IT operation management, security services and storage management services.
  - **PaaS** includes delivery of application infrastructure capabilities — including the runtime and development-time components — as public cloud services.

NOTE: Private cloud implementations are not included in this category. Third party offerings involving single tenant solutions should be included in the Outsourcing category. Internally maintained infrastructure cloud solutions are included in the Hardware and Software categories.

This measure can be helpful in adding context to the IT investment strategy from a sourcing perspective, in terms of accounting-based resources that may be insourced (for example, IT hardware, software, personnel and occupancy/facilities costs) versus services delivered by a third party (for example, outsourced services and data/voice transmission costs). As an organization increases or decreases the level of third-party/outsourced services, it may find an inverse effect in its associated personnel, hardware and/or software expenditures, depending on the scope of third-party services retained and on business requirements. The cyclical nature of capital investments in IT hardware and software may also play a significant role in an organization's IT spending outlay during a given year.

The trend towards public cloud continued as the percentage of IT spending on SaaS, and IaaS and other public cloud services increased from 5% in 2017, to 8% in 2018. The increase in public cloud appears to be coming from personnel costs (which decreased from 40% in 2014 to 36% in 2018 and hardware which decreased from 16% in 2014 to 13% in 2018. It is interesting to note that software has remained relatively flat (21% in 2014 to 22% in 2018), even though 5% of total IT spending is now related to SaaS. This may indicate that increase in SaaS is having a bigger effect on hardware costs than it is on traditional software license fees.

Figure 30. Distribution of IT Spending by Asset Category, 2017 - 2018

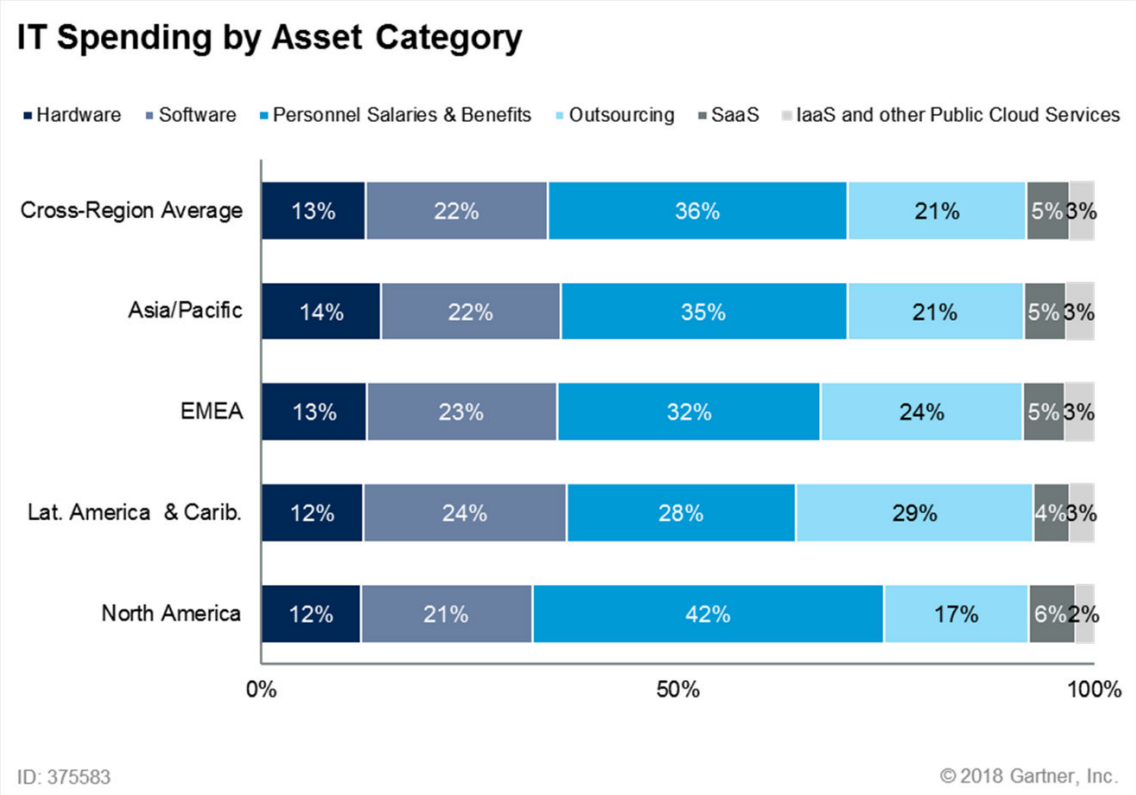


Source: Gartner IT Key Metrics Data (December 2018)

Note: For 2018 a breakout of SaaS and IaaS and other public cloud services was available and is shown. For 2017 those two categories were combined into “public cloud”. Prior to 2017 all public cloud was included as part of the outsourcing category.

Figure 31 indicates the percentage of IT spending allocated to hardware and software are broadly similar, however the split between personnel costs and outsourcing varies with North America spending more on internal personnel and less on outsourcing, and Latin America spending more on outsourcing than personnel. Regionally too there was also an increase in the public cloud percentage compared to 2017.

Figure 31. Distribution of IT Spending by Asset Category, by Region, 2018



Source: Gartner IT Key Metrics Data (December 2018)

Distribution of IT Spending by IT Functional Area, 2014 to 2018

The distribution of IT spending by IT functional area provides a view of key IT resource consumption in the context of the overall IT portfolio:

The distribution of IT spending into these categories helps to define the relative level of IT resources required per year to support the technology environment portfolio. This is often leveraged in tandem with IT resource planning exercises, wherein spending and staff resource allocations can be viewed in terms of IT infrastructure (data center, end-user computing, IT service desk, voice and data network) versus applications (application development and application support) versus IT overhead (IT management, IT finance and IT administration).

While this measure is helpful in identifying relative volumes of IT resource consumption by IT functional area, as compared to industry, it does not aid in identifying whether resources are being leveraged in a cost-effective or productive manner. For more information and ITKMD research focused on individual IT functional area cost efficiency and staff productivity, see the IT Key Metrics Data: Key Infrastructure Measures and IT Key Metrics Data: Key Applications Measures documents.

Definitions for the IT Functional Area Framework are included below.

Change of Methodology

During 2017 Gartner changed the approach for calculating the distribution of IT spending by IT functional area, moving from an “accounting view” (based on operational expenses plus depreciation and amortization and excluding capital) to a “cash flow” view (based on capital and operational expenses excluding depreciation and amortization). This brings the approach into line with how the IT spending calculations are made for the other metrics in this report.

However the historical information for the years 2014 to 2016 in Figure 32 have not been converted and are still based on an “accounting view,” as Gartner believes that in the long run these two methods produce similar results.

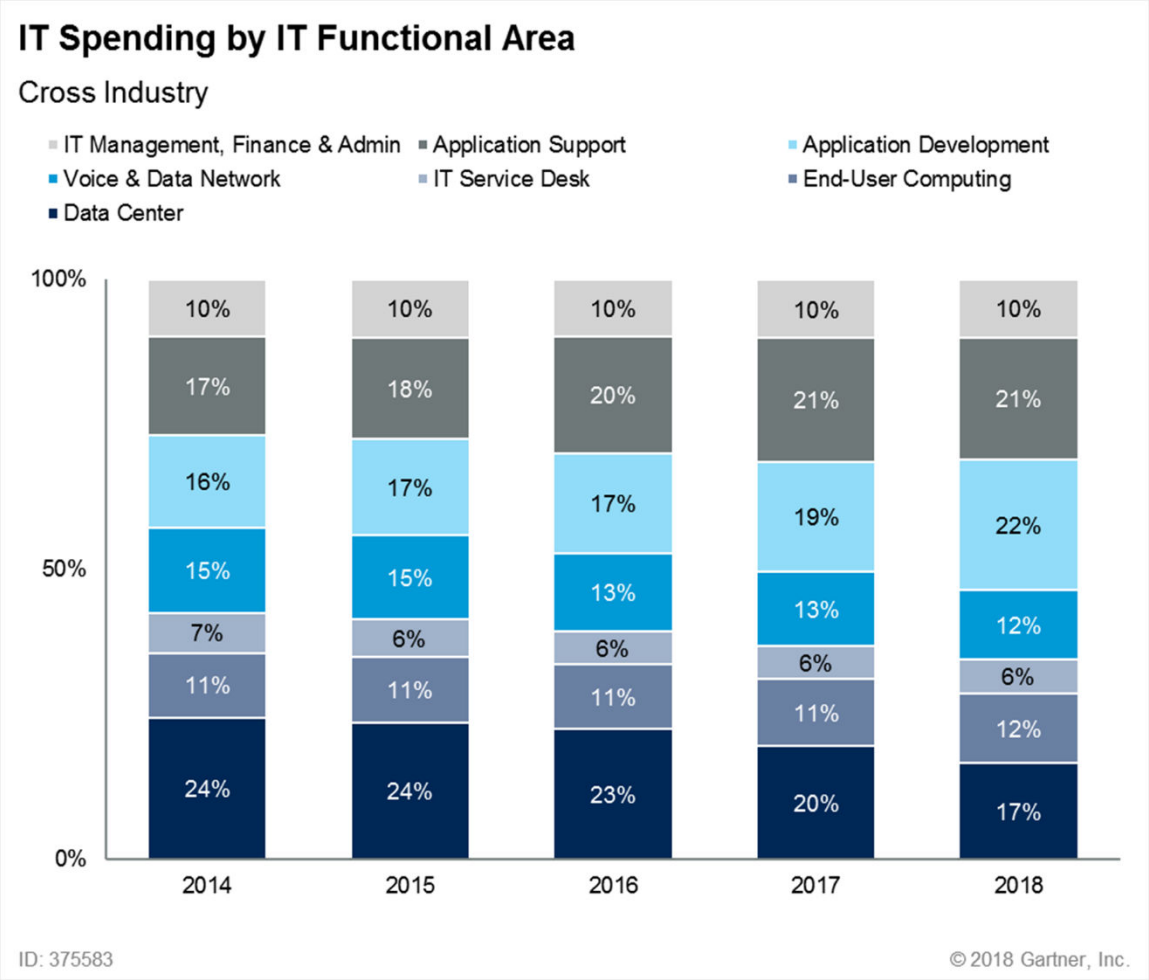
We have also collapsed the number of functional areas, combining Voice and Data Networks into a single category, and combining IT Management and Finance and Administration together.

IT Spending by IT Functional Area

On average, in 2018, more than 40% of IT spending was applications related, about 50% was infrastructure related, and the remaining 10% was dedicated to IT management, finance and administration activities.

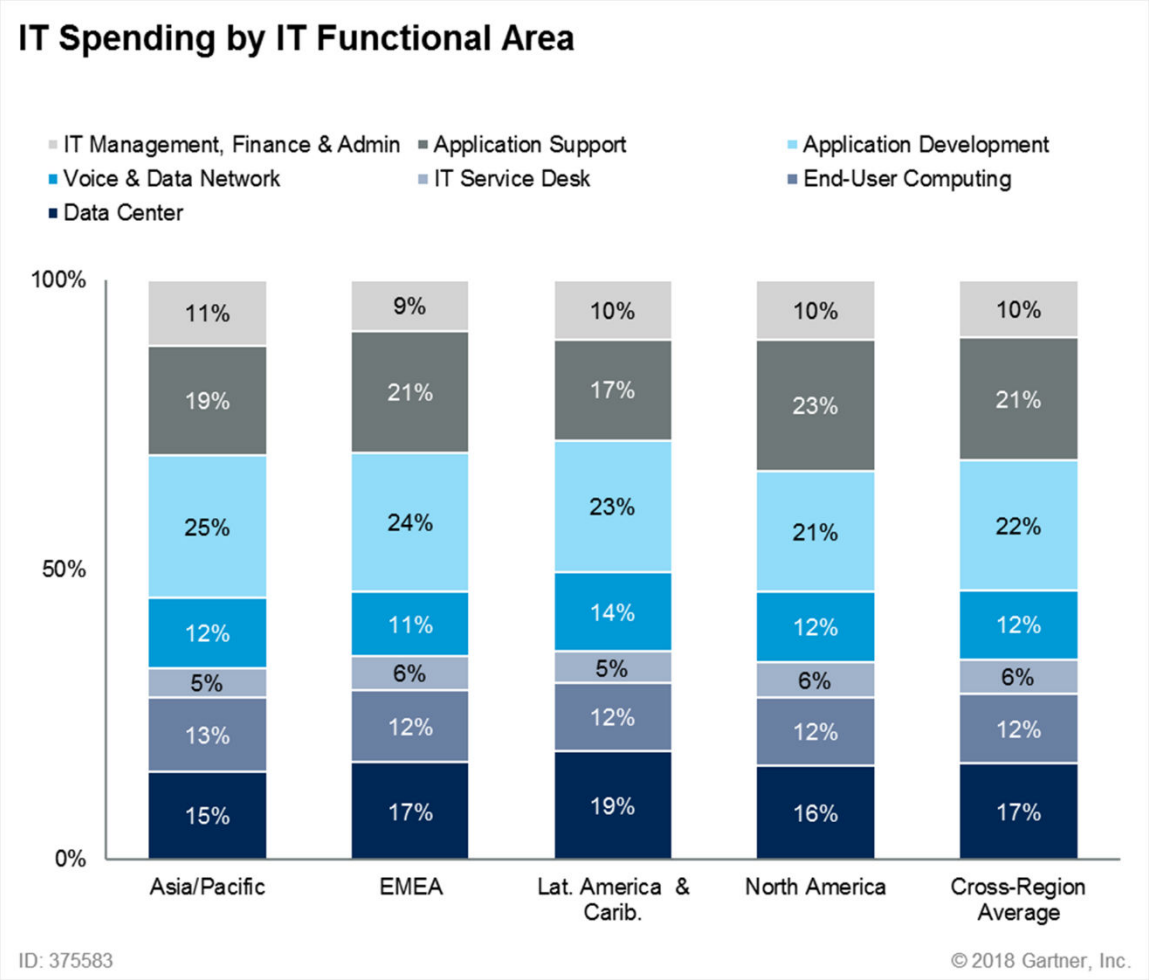
Application development and support spending has continued to take-up an increasing percentage of the overall distribution, while data center spending is decreasing and no longer takes up the largest proportion (see Figure 32).This is shift is again likely to be connected to a move towards more public cloud services.

Figure 32. Distribution of IT Spending by IT Functional Area, 2014 to 2018



Source: Gartner IT Key Metrics Data (December 2018)

Figure 33. Distribution of IT Spending by IT Functional Area, by Region, 2018



Source: Gartner IT Key Metrics Data (December 2018)

To better understand IT functional area cost-efficiency levels, Gartner recommends evaluating individual IT functional area annual costs compared with the workload supported, within the context of service levels, complexity, demand and scale. For more information on cost measures by IT functional area, see Gartner's various IT Key Metrics Data: Key Infrastructure Measures research (which is cited throughout this report and in the IT Functional Area Framework section).

IT Staffing Trends

As we have seen in preceding figures, internal staff typically represents more than one-third of the overall IT investment, which demonstrates the considerable human component of the IT portfolio. As such, it is critical for organizations to understand whether they are staffed adequately, whether their human resources are effective and whether they are sufficiently trained and motivated to meet changing business needs. The following metrics provide a broad view of IT staffing levels among the organizations we studied.

IT Full-Time Equivalents as a Percent of Employees

IT FTEs as a percent of employees is a key measure of IT support and IT intensity from a human capital perspective.

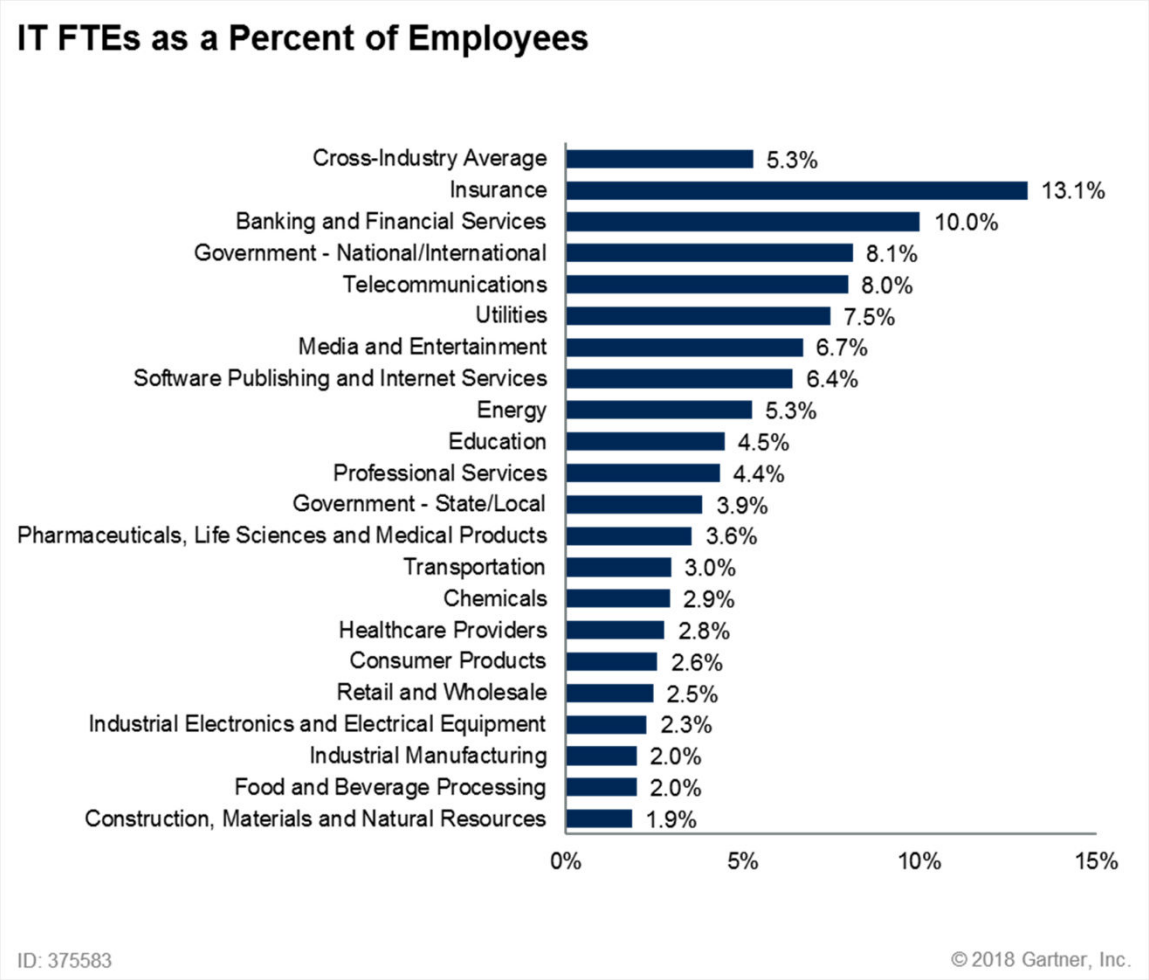
We define an IT FTE as follows:

> "An IT FTE represents the logical staff to support functions performed by the physical staff, measured in calendar time. This includes all staffing levels within the organization, from managers and project leaders to daily operations personnel. This also includes insourced FTEs and contract FTEs. However, this excludes the staff of a third-party vendor (for example, IT outsourcing), which is not operationally managed by the in-house staff, but rather is managed by the vendor."

Understanding the relative level of IT staff dedicated to supporting the business can also assist in identifying whether the staff size is appropriate. This should be considered within the context of the overall enterprise sourcing strategy and future-state objectives. Variables to consider in tandem with this metric include IT staffing distribution, contract versus insourced FTEs, and IT outsourcing and public cloud spending as a percent of IT spending, as well as the enterprise sourcing strategy — Does the total employee count accurately represent the organization's workforce that is supported by IT? Do you have the ability to track the total number of internal users supported by IT?

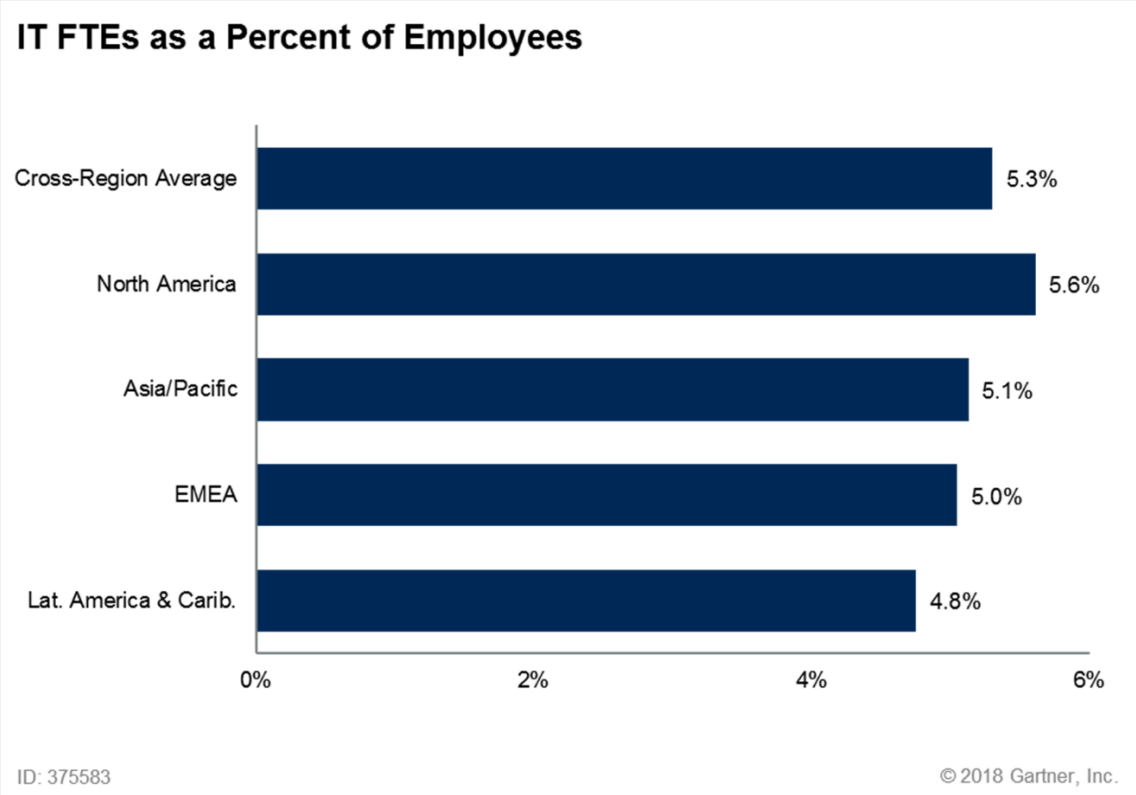
Similar to the top level IT spending metrics, the IT FTEs as a percent of employees tend to be higher for IT intensive industries such as Insurance and banking and financial services.

Figure 34. IT FTEs as a Percent of Employees, by Industry, 2018



Source: Gartner IT Key Metrics Data (December 2018)

Figure 35. IT FTEs as a Percent of Employees, by Region, 2018



Source: Gartner IT Key Metrics Data (December 2018)

Distribution of IT FTEs: Insourced Versus Contractor, 2018

The distribution of IT FTEs (insourced versus contractor) can help provide a view of the IT staffing strategy.

Insourced IT FTEs are defined as:

> "FTEs who are employed by the IT organization (excluding contractors and consultants). These include all full-time and part-time employees supporting the IT environment, as defined by IT spending/budget."

Contract IT FTEs are defined as:

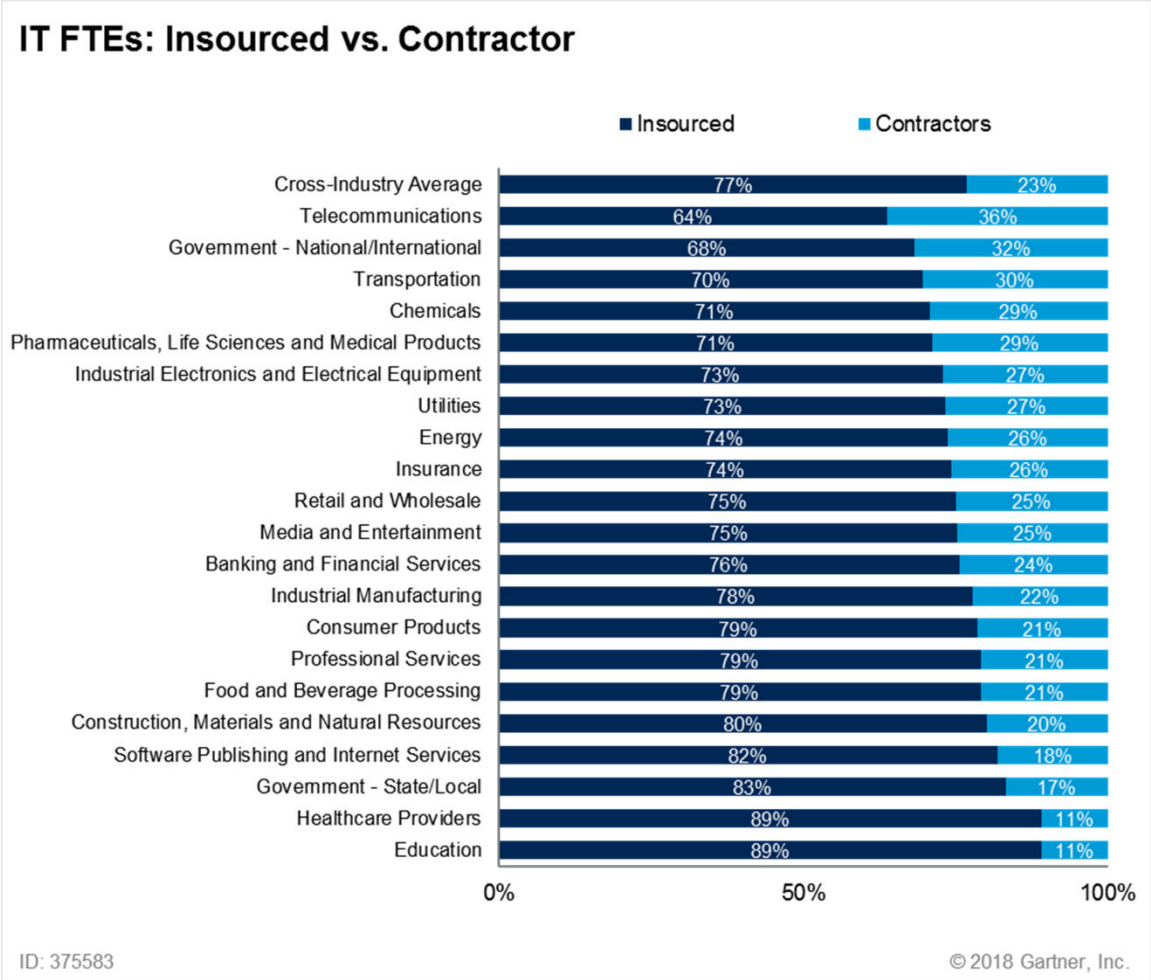
> "Contract FTEs (contractors) who are supplemental to your staff and are "operationally" managed by the in-house staff. These include all full-time, part-time and temporary contractors supporting the IT environment, as defined by IT spending/budget."

IT contract labor or contractor usage can be an effective approach to maintaining flexibility and agility when business conditions are changing. However, keeping contractors for extended periods can be costly and limit process standardization.

For more information, see "Ten Best Practices When Using External Staff Augmentation, Contractor or Temporary IT Resources."

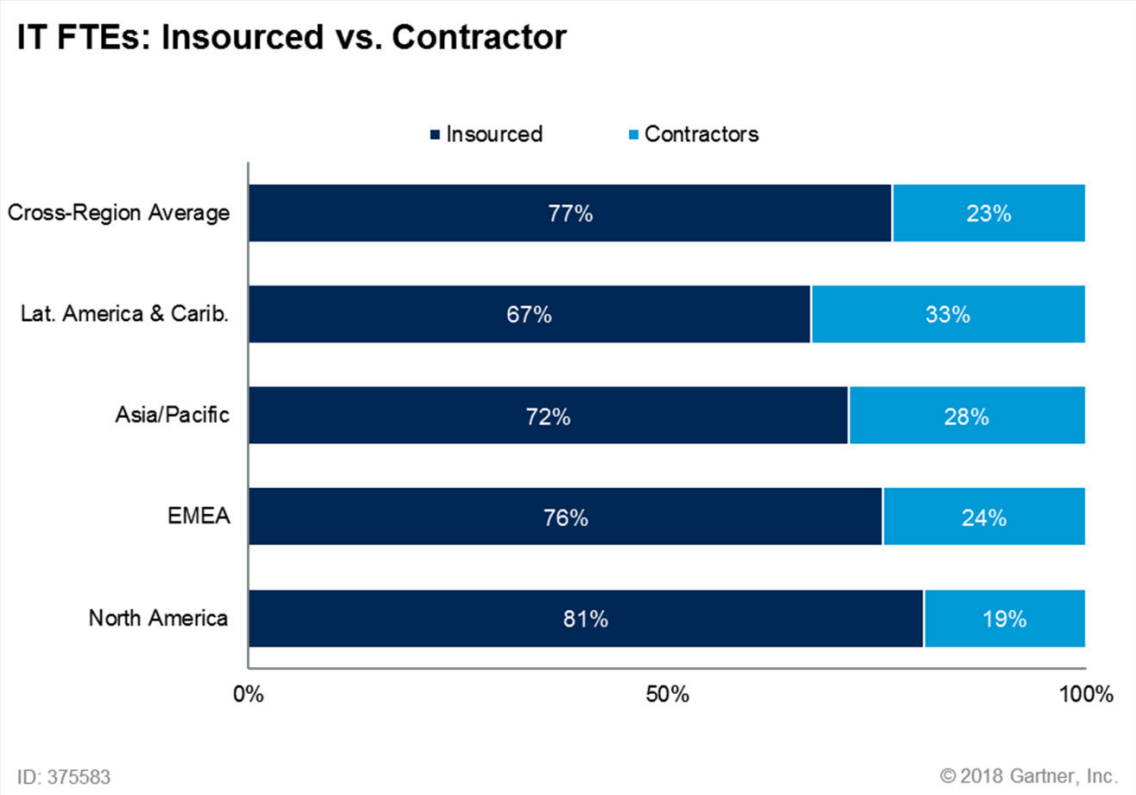
The use of contract labor should be driven by a need for flexibility and access to specific skills. However some industries such as government – national/international tend to rely on contract labor because of budgeting practices that discourage hiring full time staff. In other industries such as education and healthcare, cultural factors and a perception of having very specific business requirements lead toward a lower percentage of IT contractors in place.

Figure 36. Distribution of IT FTEs: Insourced Versus Contractor, by Industry, 2018



Source: Gartner IT Key Metrics Data (December 2018)

Figure 37. Distribution of IT FTEs: Insourced Versus Contractor, by Region, 2018



Source: Gartner IT Key Metrics Data (December 2018)

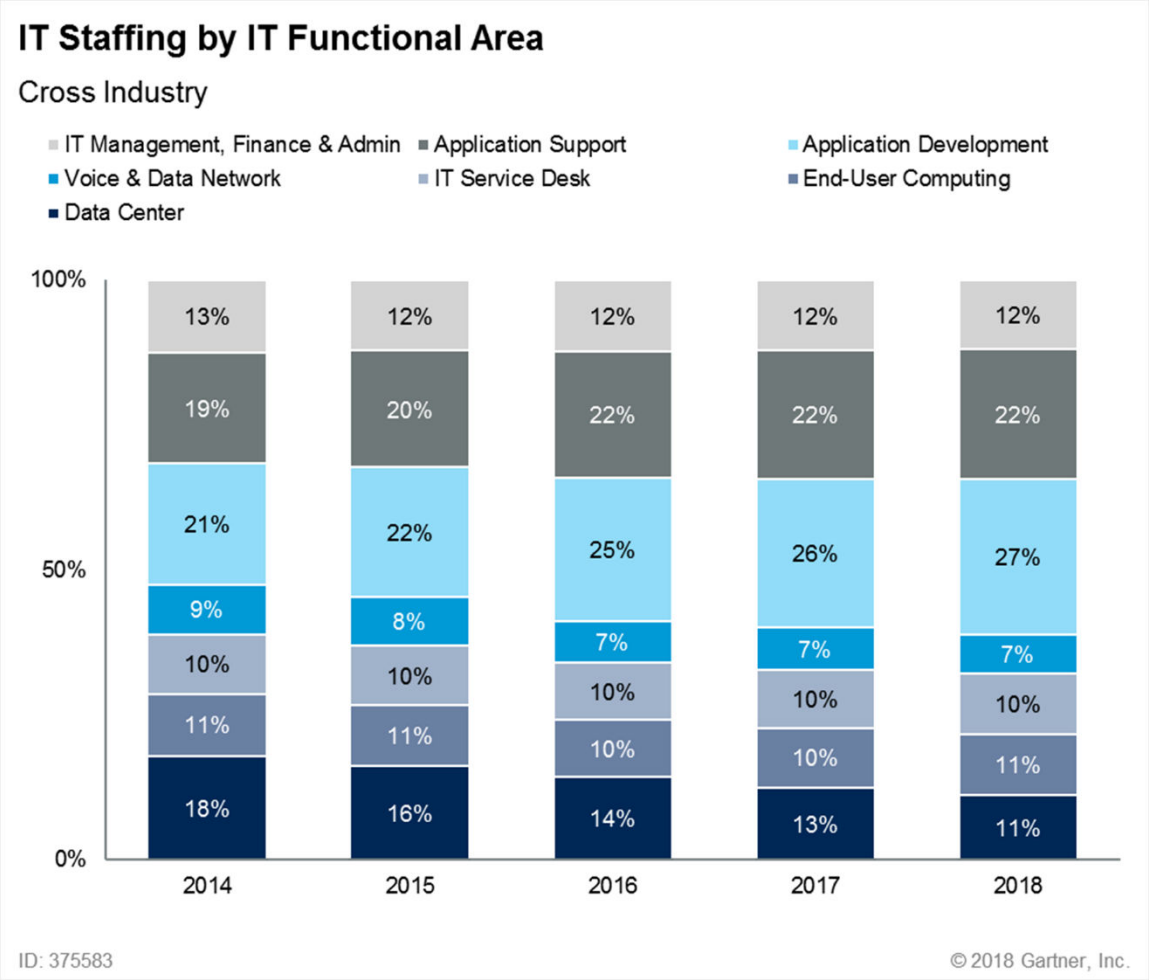
Distribution of IT Staffing by IT Functional Area

The distribution of IT staffing by IT functional area provides a view of key IT resource consumption in the context of the overall IT portfolio. Definitions for the IT Functional Area Framework are below.

By viewing human resources (IT FTEs) within the context of the total portfolio, organizations are able to identify which environment is the most labor-intensive as a percent of the IT labor pool. Typically, application activities (development and support) demand the most resources from both cost and staffing perspectives. The degree to which an organization outsources should be considered alongside such staffing metrics.

Similar to the IT spending functional area distribution we see increase in the percentage of staffing involved in application development and application support, and a decrease in the percentage in the data center.

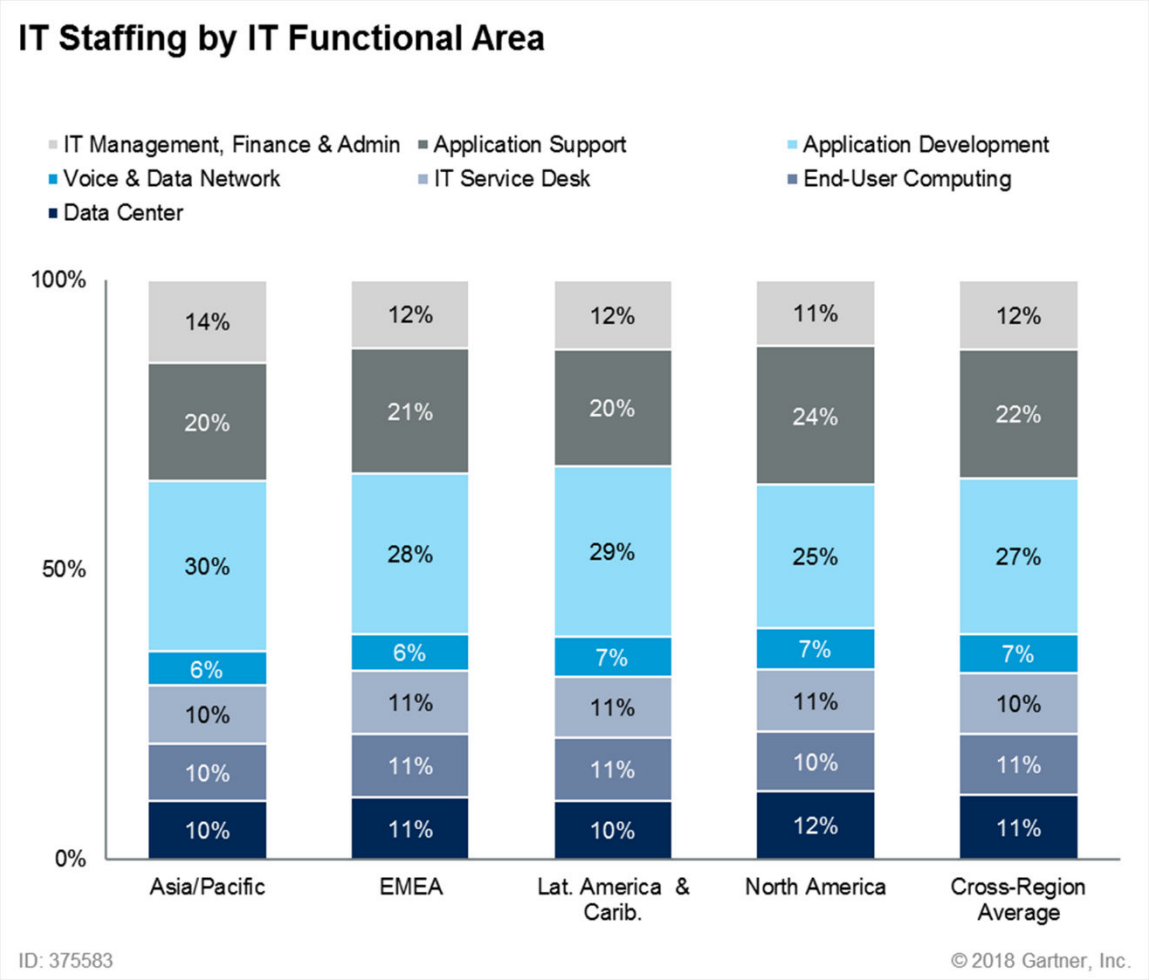
Figure 38. Distribution of IT Staffing by IT Functional Area, 2014 to 2018



Source: Gartner IT Key Metrics Data (December 2018)

Regionally, Asia/Pacific has the highest percentage of staff involved in application development, while North America has the highest percentage in application support. Again, Asia/Pacific has a higher percentage in IT Management Finance and Administration.

Figure 39. Distribution of IT Staffing by IT Functional Area, by Region, 2018



Source: Gartner IT Key Metrics Data (December 2018)

To better understand IT functional area staff productivity levels, Gartner recommends evaluating individual IT functional area staffing levels compared with the workload supported, within the context of service levels, complexity, demand and scale. For more information on productivity measures by IT functional area, see Gartner's various IT Key Metrics Data: Key Infrastructure Measures, research (which is cited throughout this report and in the IT Functional Area Framework section).

IT Functional Area Framework

The following sections provide guidance on how to count spending and FTE numbers, as defined by the scope of the IT functional area framework/chart of accounts. This includes annual capital plus operational (cash flow) spending associated with the purchase, lease, operation, and maintenance of hardware, software, connectivity, disaster recovery, occupancy/facilities and personnel to support the environment as defined below.

Data Center

The scope of the Data Center function includes

- Windows, Linux, Unix, and other standard server environments
- Provisioning and management of all data storage within an enterprise
- IBM (or other) mainframe systems running z/OS including subsequent or prior models operating on both general-purpose engines and specialty engines such as zAAP, zIIP and IFL.
- Data Center facilities.

Hardware

- Processing Devices: Includes all hardware in server platform configurations, including internal disk storage, processors, memory, cards, etc.
- Client Devices: This includes the equipment used by the operations staff to support the environment (e.g., desktops, laptops, tablets).
- Storage Controllers, Storage Servers: All dedicated storage hardware devices including controllers, servers, disk arrays, tape libraries, optical jukeboxes.
- Offline Supplies (Media): Portable media used to store data offline such as tapes.

Software

- Annual spending for host and virtual OS licenses, virtualization and partitioning software, utilities, databases, middleware, content/document management search engines, messaging, communications (TCP/IP, FTP and host-based) and server security software.
- Annual license spending on software dedicated to managing the storage systems. This includes creation and setup, storage maintenance, reporting, security, monitoring, backup/restore, archival, replication, media handling and data migration/tiering.

Connectivity

- Intra-data-center connectivity: This typically includes routers, switches, load balancers, controllers and appliances. Data center communication networks are dedicated networks that are segregated or isolated from the general-purpose LANs or WANs. General-purpose or shared network costs are excluded from the data center and should be allocated to the voice and data network category.
- Inter-data-center connectivity: This typically includes the transmission spending and hardware spending for the fiber, used and unused (dark fiber), and the switches and controllers. Data center remote communication networks are dedicated networks that are segregated or isolated from the general-purpose LAN or WAN. General-purpose or shared network costs are excluded from the data center and should be allocated to the data network.

Disaster Recovery

- Disaster Recovery Contracts (Compute and Communications) for Hot Sites (Shell facilities), Dedicated Hardware, Software, and Connectivity.
- Spending on hardware, software, connectivity, facilities and contracts specifically dedicated to disaster recovery storage management.

Facilities/Occupancy

- Occupancy spending should include fully burdened spending for the non-data center floor space being used by the staff supporting the enterprise computing and storage environment under analysis. Some examples include office space, furniture, electricity, maintenance, property taxes, security and office supplies.
- Facilities spending includes management of the physical data center premises, and other facilities and services associated with the premises such as furniture, power supply, heat management, climatization services, access security, floor space, office space, design and consulting.
- Building and Maintenance
  - Utilities — electricity, water and gas for all data center equipment.
  - Mechanical and Electrical Systems — air conditioning (CRACS), chillers, humidifiers, fans and associated piping and ducting, power distribution units (PDUs), uninterruptible power supply (UPS), backup generators/batteries, cabling and electrical conduits, fire prevention and suppression, building management systems (BMS), data center infrastructure management (DCIM), lighting, heating and physical/logical security systems.

Personnel

- Operations/maintenance, engineering technical services, planning and process management, service administration, management and administration, and facilities management.

For more detailed information, see "IT Key Metrics Data 2019: Key Infrastructure Measures: Windows Server Analysis: Current Year," "IT Key Metrics Data 2019: Key Infrastructure Measures: Linux x86 Server Analysis: Current Year," "IT Key Metrics Data 2019: Key Infrastructure Measures: Unix Server Analysis: Current Year," "IT Key Metrics Data 2019: Key Infrastructure Measures: Mainframe Analysis: Current Year" and "IT Key Metrics Data 2019: Key Infrastructure Measures: Storage Analysis: Current Year." (Access to these documents is dependent on your level of Gartner subscription.)

End-User Computing

The scope of the End-User Computing function includes

Hardware

- Personal Computing Devices: Desktop, laptop, thin client, and tablet.
- Other Client Devices: Smartphones and other handheld devices.

Transmission spending for smartphones and tablets are excluded and should be allocated to the voice and data network category.

- Peripheral hardware: Personal and shared printers as well as multifunctional printers/devices. Include costs for toner cartridges but exclude other consumables such as drum costs, ribbons, paper, and printer maintenance kits.
- IT management hardware: This encompasses hardware that primarily supports an IT process, not a business or user process. Examples are test and training devices, servers hosting network and system management (NSM) or asset management software, and devices used by the IT staff supporting the end-user computing environment. This also includes supporting a hosted virtual desktop (HVD) installation.

Software

- User client software.
- Personal productivity and database: This includes new word processors, spreadsheets, presentation packages, personal databases and other personal productivity software executing on client systems. It also includes upgrades.
- Messaging and groupware: This includes new and upgraded email, groupware and collaboration software.
- IT Management Software: This includes IT software that is used exclusively for IT functions including network, systems, storage and asset management, training and computer-based training (CBT) software as well as security software (antivirus, personal firewall, encryption, etc.) as well as mobile device management which offers software distribution, policy management, inventory management, security management and service management for smartphones and media tablets. This also includes HVD implementations comprised of dedicated server virtualization software to host desktop software (as a server workload), brokering/session management software to connect users to their desktop environments, and tools for managing the provisioning and maintenance (e.g., updates and patches) of the virtual desktop software stack.

Disaster Recovery

- Annual spending on hardware, software, connectivity, occupancy and contracts specifically dedicated to disaster recovery for end-user computing.

Occupancy

- Occupancy spending should include fully burdened costs for the facilities being used by the staff supporting the end-user computing environment. Some examples include office space, furniture, electricity, maintenance, property taxes, security and office supplies.

Personnel

- Operations/maintenance, engineering technical services, planning and process management, service administration, management and administration.

For more detailed information, see "IT Key Metrics Data 2019: Key Infrastructure Measures: End-User Computing Analysis: Current Year." (Access to this document is dependent on your level of Gartner subscription.)

IT Service Desk

The scope of the IT Service Desk function includes

Hardware

- PBX, ACD, interactive voice response, computer-telephony integration, IT service desk end-user computing devices, and IT service desk application servers.

Software

- This includes all software that is necessary to operate the IT service desk, such as expert knowledge tools, problem management tools, quality monitoring, self-service, workforce management software, workflow management software and IT service desk management portal software.

Occupancy

- Occupancy spending should include fully burdened costs for the facilities being used by the staff supporting the IT service desk environment. Some examples include office space, furniture, electricity, maintenance, property taxes, security and office supplies.

Transmission

- Includes inbound 800 service, dedicated trunking, local service, outbound long distance, internet access (for example, IT service desk portal) and networking between IT service desks.

Disaster Recovery

- Annual spending on hardware, software, connectivity, occupancy and contracts specifically dedicated to disaster recovery for IT service desk.

Personnel

- IT service desk agents, operations/maintenance, engineering technical services, planning and process management, service administration, management and administration.

For more detailed information, see "IT Key Metrics Data 2019: Key Infrastructure Measures: IT Service Desk Analysis: Current Year." (Access to this document is dependent on your level of Gartner subscription.)

Voice and Data Network

The scope of the Voice and Data Network function includes

- Wide-Area Voice Network (WAV): Traditional Outbound Long Distance and Inbound (Toll Free) voice services. This includes mobile voice.
- Voice Premise Technology (VPT) — Local voice services including voice mail and all move/add/change activity associated with local voice equipment (e.g., switches, circuits and handsets), and local (i.e., not Long Distance) calling. These assessments can encompass PBX/PABX and VoIP technologies.
- Wide-Area Network (WAN): Connectivity and transmission of business-critical data between enterprise locations and business partners. This includes mobile data.
- Local-Area Network (LAN): Accounts for the provisioning of communications and connectivity to critical business systems within enterprise sites and campuses.

  Note: Spending associated with permanent building cabling, horizontal and vertical, are excluded. Likewise, spending for any inter-building cabling (copper and/or fiber) that would be found on a campus are also excluded.
- Internet Access Services (IAS): Enterprise access to the global Internet, for the use of its personnel and for the use of its external customers to access enterprise websites.

Hardware

- Wide-area voice network hardware: Switching and routing, as well as terminating hardware. Terminating hardware includes microwave, satellite, compression, multiplexer/channel bank, PBX network interface card and channel service unit/data service unit (CSU/DSU).
- Voice premise: Telephone system equipment (such as voice switch/server and peripherals, including modules and uninterruptible power supply [UPS]), premise system phones (voice only; smartphones such as iPhone and Android-based devices are excluded and should be allocated to the end-user computing environment), voice mail hardware (for example, processors and storage) move add change (MAC) materials.
- Security hardware: Dedicated data network firewall hardware/servers, intrusion/detection servers and devices, as well as encryption hardware.
- Switching, routing and wireless hardware, including switches and routers, multiplexers, satellite equipment, boundary (branch) routers, backbone routers and bridges, and wireless access points.
- Other dedicated data network hardware, including Domain Name System (DNS) and Dynamic Host Configuration Protocol (DHCP) servers, optimization equipment such as Internet load-balancing hardware, MAC hardware and MAC cable (closet to desktop).
- IT management (network operations center [NOC]): This includes hardware that is located within a client's NOC and is used to support a client's centrally managed infrastructure/network. This

includes client devices (PCs on NOC desktops) as well as servers (NOC), located within the NOC or elsewhere, but used primarily by the NOC.

Software

- Switch/voice server and peripherals (e.g., automatic call distribution [ACD], voice response unit [VRU]) and voice mail software spending.
- IT management (NOC): Software used by the NOC primarily to support/manage a client's voice and data networks.
- Security software: Dedicated data network firewall software, intrusion/detection software as well as encryption software.
- IT Management (network operations center (NOC)) software: All Network Systems Management software spending related to the NOC's support of the infrastructure/network.

Transmission

- Includes all outbound and inbound transmission spending. It also includes the annual spending on local central office lines (where applicable) as well as cellular (mobile) voice only transmission spending.
- Annual data network transmission spending, such as carrier digital services including Frame Relay access, ports and PVCs (Permanent Virtual Circuits), ATM (Asynchronous Transfer Mode) access, ports and PVCs, MPLS (Multiprotocol Label Switching) access, ports, and CARs (Committed Access Rates) which also includes specific charges for Quality of Service (QoS) commitments, sometimes referred to as traffic shaping, T3/E3, dial backup service, Synchronous Optical Network (SONET), metropolitan Ethernet, and dark fiber, as well as annual spending on circuits connected to the Internet service provider and cellular (mobile) data transmission spending.

Disaster Recovery

- Disaster recovery contracts (communications) for hot sites (shell facilities), dedicated hardware, software, and connectivity.

Occupancy (For Personnel Only)

- Occupancy spending should include fully burdened spending for the facilities being used by the staff supporting the voice and data network function. Some examples would include office space, furniture, electricity, maintenance, property taxes, security and office supplies. Occupancy for hardware (for example, closet space) is specifically excluded (that is, occupancy spending should apply only to the people supporting a client's voice and data networks).

Personnel

- Operations/maintenance, engineering technical services, planning and process management, service administration, management and administration.

For more detailed information, see "IT Key Metrics Data 2019: Key Infrastructure Measures: Voice and Data Network Analysis: Current Year," "IT Key Metrics Data 2019: Key Infrastructure Measures: Local-Area Data Network Analysis: Current Year," "IT Key Metrics Data 2019: Key Infrastructure Measures: Wide-Area Data Network Analysis: Current Year," "IT Key Metrics Data 2019: Key Infrastructure Measures: Voice Premise Technology Analysis: Current Year." and "IT Key Metrics Data 2019: Key Infrastructure Measures: Wide-Area Voice Network Analysis: Current Year," (Access to these documents is dependent on your level of Gartner subscription.)

Applications

The scope of the Applications function includes

Application Development

- New code for a new application and functional enhancements to the current code that take more than two person-weeks, or that typically add eight function points or more. A "functional enhancement" is defined as "a change made for a user that allows additional capabilities (from a business point of view) that were not there before. In some environments, major enhancements can actually be added in less than two person-weeks. If this is the case, and eight function points or more are added (about 800 lines of COBOL or 300 lines of a database language), then this is still categorized as development.

Application Support

- Bug fixes of any size or duration, maintenance of hard-coded data or tables (including field size changes) embedded within the programs (any size or duration), and functional enhancements to current code that take two person-weeks or less and typically add fewer than eight function points, or any project that produces no new business functionality for the user.
- A "functional enhancement" is defined as "a change made for a user that allows additional capabilities (from a business point of view) that were not there before." In some environments, major enhancements can actually be added in less than two person-weeks. If this is the case, and eight function points or more are added (about 800 lines of COBOL or 300 lines of a database language), then this is categorized as development rather than support.

Hardware

- This includes only hardware (mainframes, servers, end-user computing devices) used by the application development or support staff members to do their jobs (that is, client devices as well as servers and a portion of the mainframe used for application development and testing). This excludes end-user or production hardware.

Software

- Development and support software required by the application development and support staff members to do their jobs. It may include the languages/compilers/databases, development/testing tools and IT management software tools, such as project estimators and project schedulers.

- Business functionality software: For application support, this includes the maintenance cost of off-the-shelf vendor packages or SaaS fees.
- PaaS if used as a development platform.

Occupancy

- Fully burdened spending for the facilities used by the development or support staff included in this analysis view. Some examples would include office space, furniture, electricity, maintenance, property taxes, security and office supplies.

Personnel

- Development, logical database analyst, quality assurance & testing, IT process management, project & program management, services administration, management and administration.
- Application development: This includes staff involved in developing new applications, enhancing existing applications, installing new packages and installing major functional enhancements to existing packages.
- Application support: This includes staff involved in supporting applications that exist within the current portfolio. It also includes personnel who are responsible for fixing programming problems uncovered when applications are running in production. It does not include any personnel who are responsible for running the production applications. If an upgrade for a packaged application primarily contains fixes for existing problems, then the efforts involved in installing such a maintenance upgrade are included in application support.

For more detailed information, see "IT Key Metrics Data 2019: Key Applications Measures: Cost and Staff Profile: Current Year," "IT Key Metrics Data 2019: Key Applications Measures: Application Development: Current Year" and "IT Key Metrics Data 2019: Key Applications Measures: Application Support: Current Year." (Access to these documents is dependent on your level of Gartner subscription.)

Corporate IT Management, Finance, and Administration

Only include functions that are at a level within the IT organization that, after best effort, cannot be allocated to an IT functional area.

Office of the CIO/CTO

- This includes the "C-level" IT management, including the CIO and CTO functions. Also included here are the direct reports of the CIO, who spend the majority of their time providing enterprise wide support other than the functions outlined below (that is, special projects).

IT Human Resources

- This includes resources dedicated to human resource issues surrounding the recruiting and retention of IT staff.

IT Marketing

- This includes resources dedicated to marketing the capabilities of the IT organization to the business units.

Technology Planning and Process Management

- This includes activities related to the planning for and management of current and future technology needs, and the establishment of policies and processes relating to technology. This also includes, but is not limited to, systems research, product management, technology evaluation and purchase decision making, the establishment of processes surrounding security and virus protection, and business continuity/recovery.

Disaster Recovery

- This includes resources dedicated to planning, testing and implementing contingency procedures across all IT functions. This also includes the staff dedicated to safeguarding the enterprise's ability to continue operations of vital business functions following physical damage or other catastrophes that impact business facilities. Responsibilities include:
  - Maintaining disaster recovery documentation
  - Negotiating contingency site arrangements and serving as liaison with the vendor
  - Managing off-site data retention.

Security

- This includes resources that oversee the development of standards and procedures for ensuring overall network and systems integrity.

IT Administration

- This includes direct administrative and clerical support to enterprise-level IT. Positions include secretary, receptionist and administrative assistant.

Budget and Chargeback

- This area establishes the overall IT budget, monitors actual expenses versus the budget, arranges financing for purchases and performs financial reporting to other enterprise areas. Staff members also handle the operation of the chargeback system. Positions include financial analyst and chargeback administrator.

Asset Management

- Tracking: This area provides the administrative support for tracking systems and system components. It accounts for labor and contract spending for managing depreciation records and lease contracts, performing asset inventories (physical or automatic management), asset identification and tracking, asset database management, change recording and reconciliation. It also includes the creation and maintenance of an up-to-date record of installations, moves, adds, changes, removals and final disposal of all assets (for example, hardware, software and circuits). The record contains information for locating, assessing, auditing, troubleshooting,

counting and assigning assets, or performing other technical and business functions without the need to repeatedly visit the asset location or reassemble data records. It also includes the determination of an asset's useful life, including planning for the installation, upgrade, and removal/disposal of the asset and executing the plan.

- Procurement: This area solicits bids, negotiates purchasing agreements, establishes purchase orders, validates vendors' bills, coordinates with accounts payable for payments and handles contract administration.

Quality Assurance

- This includes staff responsibility for monitoring, tracking and recommending solutions for improving the content and delivery of the overall services provided by IT.

Training

- This refers to the primary source for the delivery of training within the IT organization and for end users in the business units. This area may also prepare the training materials, evaluate employee skills and assist in the creation of custom training programs for the organization.

Conclusions

A successful IT performance measurement program communicates metrics that are important to a target audience. This remains true when communicating IT investments to the business. The metrics and benchmarks that Gartner has identified here provide a high-level view of current trends in IT by industry. They also reveal trends in business alignment, staffing, technology and outsourcing. They can be used to assist in communicating alignment with the business and in evaluating targets in key technology areas. They provide context for key business decisions and internal performance measures.

It is important to understand that the published averages are not targets, and decisions of "good" or "bad" performance should not be based on these metrics. They are indicative reference points from which to view current performance and investment levels to help you identify differences that could merit further analysis. Articulating why your organization is higher or lower than these metrics is the first step in better business alignment and the communication of IT's impact on business performance.

For more detailed metrics focused on IT infrastructure cost and performance, consult Gartner's various IT Key Metrics Data: Key Infrastructure Measures research, which can help provide more insight into IT-centric cost-efficiency and productivity metrics.

For more detailed metrics focused on IT application spending, staffing and project measures, consult Gartner's IT Key Metrics Data: Key Applications Measures research, which can help provide more insight into total application development versus support metrics.

ITKMD is a Gartner Benchmark Analytics solution that delivers indicative IT metrics in a published format as directional insight for IT organizations. This solution represents a subset of the metrics and prescriptive benchmark analysis capability that is available through Gartner Benchmark Analytics. For ongoing and more targeted analyses, Gartner Benchmark Analytics provides clients

with in-depth, personalized benchmarking and customized assessments. These prescriptive, client-focused engagements are structured to identify technology performance strengths, to prioritize opportunities for IT and business optimization, and to assist in communicating IT's role in creating business value through strategy enablement and process improvement.

Related IT Key Metrics Data Research

This research is part of a set of Gartner Benchmark Analytics research pieces.

Depending on your subscription level for Gartner services, some clients have access to the Gartner ITKMD publication series from gartner.com, select "Explore," "Metrics & Tools," and "IT Key Metrics Data."

For detailed information and metrics specific to each of the IT Key Metrics Data: Key Industry Measures, see individual documents outlined in Table 1 above, or review "IT Key Metrics Data 2019: Index of Published Documents and Metrics" for a comprehensive list of all available IT Key Metrics Data 2019 research.

Table 3. ITKMD 2019: Overview Document Index

Document Title	Current Year RN#
Executive Summary	G00375583
Midsize Enterprise Executive Summary	G00375584
Resources to Review Your ITBudget Comparison Report	G00375585
Index of Published Documents and Metrics	G00375586
Demographics	G00375587
Surveys	G00375588
Frequently Asked Questions	G00375589
Glossary of Terms	G00375590
Definition of Industries	G00375591

Source: Gartner IT Key Metrics Data (December 2018)

Appendix: Exploring Gartner's Prescriptive Benchmark Analytics Capabilities

Fueled by market-leading data and actionable insights delivered by our specialist practitioners, you can make fact-based decisions to:

- **Measure** — know where you are today, understand your costs compared to peers and determine prudent ways to sustain your baseline
- **Optimize** — assess your best opportunities for improvement and nurture your growth
- **Transform** — use our comparative measurement to align with your business initiatives and demonstrate progress

Internal operations

Understand your current state and identify pertinent opportunities to propel growth by asking:

- Is your IT spending aligned with your performance and value goals?
- Can you independently prove that you are delivering value for money?
- Can you sustain your baseline costs and identify improvement areas?

Internal operations benchmarks:

- **IT and business leader budget baseline:** Cost control is a must-have for any enterprise. Leadership is increasingly focusing on value creation via employee productivity, customer retention and competitive advantage.
- **New-CIO baseline:** You need a baseline to understand what you have inherited, identify your focus areas and determine how you can measure the impact of your strategy.
- **Applications cost, staffing and productivity:** Assessing the productivity and cost-efficiency of applications enables you to demonstrate the value of IT spending and prioritize future application spending.
- **Infrastructure cost and staffing efficiency:** Understanding and comparing performance and support of IT network, storage and infrastructure allow you to present IT value and cost optimization opportunities to the business.
- **Security risk, maturity and spending:** Assess the effectiveness of your security budgets and execution to determine the investments and initiatives required to achieve acceptable risk tolerance.
- **Customer satisfaction:** Measuring IT's efficiency and effectiveness by evaluating end-user satisfaction with IT services enables you to prioritize efforts and optimally use your resources to improve future satisfaction levels.

Market perspective

Know the market competitiveness of your service contracts by asking:

- Can you prove that the terms of your contracts are good?
- Do you have the necessary market insights for contract renegotiations?
- Can you measure the performance and price of your service provider?

Market perspective benchmarks:

- **Market price baseline:** Gain valuable insight into operational efficiencies and cost reduction considerations by understanding the competitive market price ranges for services, thereby effectively demonstrating the value of IT to the business.
- **Contract price competitiveness:** Achieve the best value and efficiencies from an outsourced investment by assessing the price competitiveness of a provider's contract against comparable peers.
- **Service catalog rates:** Make informed decisions on cost efficiencies by ensuring that chargeback unit prices are aligned to the market or other internal providers.
- **Market cloud baseline:** Understand the market maturity of cloud solutions and ensure their commercial competitiveness.
- **Business effectiveness:** Demonstrate how IT's effectiveness in the execution of mission-critical priorities helps identify opportunities to better align the IT organization with the enterprise to optimize value and results.

To learn more about "Gartner Consulting's Benchmark Analytics" contact your account executive or inquiry@gartner.com.

Gartner Benchmark Analytics and Case Studies

CIO and IT Executive Benchmarks

CIO and IT executive benchmarks evaluate performance from two perspectives: a cost and maturity assessment of critical IT competencies and IT business value. For case studies and to learn more see, "For Your IT Challenges — CIO & IT Executives."

Infrastructure and Operations Benchmarks

Infrastructure and operations benchmarks create a starting point in the process of helping IT organizations identify and assess all IT performance levels. For case studies and to learn more see, "For Your IT Challenges — Infrastructure and Operations."

Applications Benchmarks

Applications benchmarks are the starting point in the process to help IT organizations identify and assess application development and support performance levels. For case studies and to learn more see, "For Your IT Challenges — Applications."

Sourcing and Vendor Relationship Benchmarks

Sourcing and vendor relationship benchmarks provide an accurate answer to the question, "Is this a good market price for the services being provided?" For case studies and to learn more see, "For Your IT Challenges — Sourcing and Vendor Relationships."

End-User Satisfaction Benchmarks

IT customer satisfaction benchmarks establish a baseline for customer satisfaction and create a roadmap that helps prioritize efforts to increase these levels. For case studies and to learn more see, "For Your IT Challenges — IT Customer Satisfaction."

IT Business Effectiveness Benchmarks

IT business effectiveness benchmarks establish a baseline for IT's effectiveness in meeting business needs and identify opportunities to better align the IT organization with the enterprise for maximum results. For case studies and to learn more see, "For Your IT Challenges — IT Business Effectiveness."

Gartner Recommended Reading

Some documents may not be available as part of your current Gartner subscription.

"Toolkit: How to Present Your IT Budget to the Board of Directors"

"Toolkit: Gartner's Top 100 IT Cost Optimization Ideas"

"Key Concepts in IT Financial Management: Budgeting, Funding, Transparency and Allocation"

“Toolkit: The Gartner Cost Value Matrix, 2018: A Framework to Measure IT Efficiency and Business Value”

“IT Cost Optimization Should Be an Ongoing Discipline”

“How CIOs Can Work Effectively With CFOs to Optimize Cost”

“Three-Year Roadmap for IT Cost Optimization”

“Gartner's Top Cost Optimization Techniques: An IT Perspective”

“Win at Cost Optimization With Nine Critical Success Factors and Their Metrics”

“Run IT as a Business Using Six Pillars of IT Financial Transparency to Drive Value”

Evidence

- This research contains relevant cross-industry averages and ranges from a subset of metrics and prescriptive engagements available through Gartner Benchmark Analytics consulting-based capabilities.
- Employee, income and revenue data is based on the most recently completed fiscal year.
- Calculations were made using worldwide observations.

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